

Project Report
on

Smart Class Room Monitoring System and Timetable Alerts Using Deep Learning

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BONAFIDE CERTIFICATE

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DECLARATION

We hereby declare that the project work entitled “**SMART CLASS ROOM MONITORING SYSTEM AND TIMETABLE ALERTS USING DEEP LEARNING**” has been done by us under the guidance of **Dr. P. K. Venkateswara Lal, Ph.D**, Department of Computer Science and Engineering. This project work has been submitted to the **Narayana College of Engineering, (Autonomous), Gudur** as partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology**.

We also declare that this project report is not copied in part or whole or otherwise plagiarized the work of other students and/or persons or any entity.

We also declare that this project report has not been submitted at any time to another Institute or University for the award of any degree.

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ABSTRACT

In many educational institutions, classrooms may remain unsupervised due to faculty unavailability, leading to disruptions and loss of learning. Manual classroom monitoring is inefficient, especially in large campuses. To overcome this, we propose a web-based smart classroom supervision system built using Python, OpenCV, face recognition libraries, Flask web framework, and Twilio SMS API.

A webcam-based face recognition module detects the presence of registered faculty members in classrooms. A simple web dashboard displays the real-time supervision status of each classroom. If a faculty member is detected, a green indicator appears next to the classroom on the dashboard. If no faculty is present for a specified time, it turns red, alerting the admin visually. Additionally, the system is linked to the faculty timetable and sends SMS reminders to teachers before their scheduled class, enhancing punctuality.

The system is designed to be scalable and can be easily deployed across multiple classrooms within an institution. It minimizes the need for manual supervision and reduces administrative workload. Furthermore, it provides quick decision-making support by offering real-time alerts and status updates. The system ensures non-intrusive monitoring without recording or surveillance, making it practical and respectful of faculty privacy.

Overall, this approach offers a cost-effective, reliable, and efficient solution for improving classroom management, ensuring continuous supervision, and enhancing the overall quality of education delivery in institutions.

TABLE OF CONTENTS

Content	Page No's
List of Tables	
List of Figures	
List of Symbols and Abbreviations	
Chapter – 1: INTRODUCTION	1-4
1.1 Problem Statement	1
1.2 Need for the Study	1
1.3 Motivation	2
1.4 Scope of the Work	2
1.5 Objectives of the Proposed Work	3
1.6 Overview of the Thesis	4
Chapter – 2: LITERATURE SURVEY	5-15
2.1 Introduction	5
2.2 Literature Survey on Face Recognition-Based Systems	6
2.3 Literature Survey on Smart Monitoring Systems	8
2.4 Literature Survey on Automated Alert Systems	9
2.5 Literature Survey on Attendance Analysis & Reporting Systems	11
2.6 Literature Survey on Role-Based Access Control	12
2.7 Comparative Analysis of Existing Systems	14
2.8 Conclusion	15
Chapter – 3: SYSTEM ANALYSIS	16-20
3.1 Overview	16
3.2 Existing System	16
3.2.1 Challenges in Existing System	17
3.3 Proposed System	17

3.3.1 System Objectives	18
3.3.2 Features of Proposed System	18
3.4 Feasibility Study	19
3.4.1 Technical Feasibility	19
3.4.2 Operational Feasibility	19
3.4.3 Economic Feasibility	19
3.5 Conclusion	20
Chapter – 4: DESIGN PROCESS	21-32
4.1 System Architecture	21
4.2 UML Diagrams	21
4.2.1 Class Diagram	22
4.2.2 Use Case Diagram	23
4.2.3 Sequence Diagram	23
4.2.4 Deployment Diagram	24
4.3 Architecture Diagram	25
4.3.1 Modules Description	26
4.3.2 Data Processing & Storage	27
4.3.3 Visualization & Analytical Module	28
4.4 Data Flow Diagram (DFD)	28
4.5 Architecture Flow	30
4.6 User Interface Design	31
4.7 System Requirements	32
4.7.1 Hardware Requirements	32
4.7.2 Software Requirements	32
Chapter – 5: METHODOLOGY	33-37
5.1 Introduction	33

5.2	Technology Description	33
5.2.1	Front-End Technologies	33
5.2.2	Back-End Technologies	34
5.2.3	Data Storage	34
5.3	System Architecture	35
5.4	Methodology Workflow	36
5.5	Advantages of Methodology	37
5.6	Limitations	37
Chapter – 6: RESULTS & DISCUSSIONS		38-48
6.1	Introduction	38
6.2	System Implementation Results	38
6.3	Performance Evaluation	45
6.4	Comparative Analysis	45
6.5	Role-Based Access Control Evaluation	46
6.6	Discussion	47
6.7	Future Improvements	48
6.8	Conclusion	48
Chapter – 7: CONCLUSION & FUTURE SCOPE		49-50
7.1	Conclusion	49
7.2	Future Scope	50
Chapter – 8: REFERNCES		51-52

LIST OF TABLES

Table No's	Title of the Table	Page No's
2.1	Comparative Analysis of Existing Systems	14-15
6.1	Comparative Analysis	45-46

LIST OF FIGURES

Figure No's	Title of the Figure	Page No's
4.2.1	UML Class Diagram	22
4.2.2	UML Use Case Diagram	23
4.2.3	UML Sequence Diagram	24
4.2.4	UML Deployment Diagram	25
4.3	Architecture Diagram	25
4.4.1	Level-0 Data Flow Diagram	29
4.4.2	Level-1 Data Flow Diagram	29
4.5	Architecture Flow	30
4.6.1	Login Page	31
4.6.2	Main Screen	31
5.3	System Architecture	35
5.4	Methodology Work Flow Diagram	36
6.2.1	Face Capture Screenshot	38
6.2.2	Checking Image Format	39
6.2.3	Face Check	39
6.2.4	App.py Capture	40
6.2.5	Home Page (Before Detection)	40
6.2.6	Home page (After Detection)	41
6.2.7	View Details (Before Detection)	41
6.2.8	View Details (After Detection)	42
6.2.9	Timetable Menu	42
6.2.10	Capture Face Menu	43
6.2.11	Faculty Data Menu	43
6.2.12	Add Faculty Button	44
6.2.13	SMS Alert (Before 5 Minutes the Class Starts)	44
6.2.14	SMS Alert (After 5 Minutes If Faculty is Not Detected)	44
6.5.1	Admin Access	46
6.5.2	Faculty Access	47

LIST OF SYMBOLS AND ABBREVIATIONS

Abbreviation	Full Form
OpenCV	OpenSource Computer Vision Library
SMS	Short Message Service
AI	Artificial Intelligence
IoT	Internet of Things
CCTV	Closed-Circuit Television
LBPH	Local Binary Patterns Histogram
CNN	Convolutional Neural Networks
RBAC	Role-Based Access Control
NIST	National Institute of Standards and Technology
ACL	Access Control List
ML	Machine Learning
UML	Unified Modelling Language
SQL	Structured Query Language
API	Application Programming Interface
RAM	Random Access Memory
HTML	Hyper Text Markup Language
CSS	Cascading Style Sheets
JSON	JavaScript Object Notation
MFA	Multi-Factor Authentication

1. INTRODUCTION

1.1 PROBLEM STATEMENT

In modern educational institutions, maintaining proper classroom supervision is essential to ensure discipline and effective learning. However, classrooms are often left unsupervised due to faculty unavailability, resulting in disruptions and reduced teaching efficiency. Traditional manual monitoring methods are time-consuming, inefficient, and impractical, especially in large campuses.

To overcome these challenges, a Smart Classroom Monitoring System using Face Recognition and Timetable Alerts is proposed. This system leverages technologies such as Python, OpenCV, face recognition algorithms, and the Flask web framework to automate classroom supervision.

A webcam installed in the classroom captures real-time images, and the system detects the presence of registered faculty members using face recognition techniques. The system is integrated with the institutional timetable and sends automated SMS reminders to faculty before their scheduled classes. A web-based dashboard provides real-time monitoring of classrooms using visual indicators.

Additionally, the system includes a weekly report feature, where administrators can analyse faculty attendance and performance over time. The system is designed to be privacy-friendly, as it does not store video recordings, ensuring ethical use of technology.

1.2 NEED FOR THE STUDY

The need for this study arises due to the following limitations in existing systems:

- Lack of automated classroom supervision systems
- Inefficiency of manual monitoring methods
- Difficulty in managing multiple classrooms simultaneously

- No real-time tracking of faculty presence
- Absence of alert systems for scheduled classes
- No analytical tools for evaluating faculty attendance

Unsupervised classrooms negatively impact learning outcomes and institutional discipline. Therefore, there is a need for an intelligent system that can automate monitoring, improve punctuality, and provide actionable insights through reports.

1.3 MOTIVATION

The motivation behind this project is to enhance classroom management using modern technologies such as artificial intelligence and web-based systems.

Key motivating factors include:

- Reducing manual workload for administrators
- Ensuring classrooms are continuously supervised
- Improving faculty punctuality through automated reminders
- Utilizing face recognition for accurate detection
- Providing data-driven insights through weekly reports
- Developing a scalable and efficient smart campus solution

This project aims to create a reliable system that not only monitors but also improves the overall teaching environment.

1.4 SCOPE OF THE WORK

The scope of this project includes the design and implementation of an automated classroom monitoring system with the following features:

- Real-time faculty detection using face recognition
- Integration with faculty timetable
- Automated SMS alert system
- Web-based dashboard for monitoring classroom status
- Generation of weekly attendance reports for faculty

The system can be extended in the future to include:

- Student attendance tracking
- AI-based behaviour analysis
- Mobile application integration
- Cloud-based data storage and analytics

1.5 OBJECTIVES OF THE PROPOSED WORK

The main objectives of the proposed system are:

- To develop an automated classroom monitoring system
- To detect faculty presence using face recognition technology
- To provide real-time monitoring through a web dashboard
- To send automated SMS alerts based on timetable schedules
- To reduce manual supervision efforts
- To ensure classrooms are not left unattended
- To generate weekly reports for analysing faculty attendance
- To design a privacy-friendly and scalable system

1.6 OVERVIEW OF THE THESIS

The report is systematically organized into multiple chapters to provide a clear understanding of the proposed system, covering theoretical concepts, design, implementation, and results. This structured approach ensures a logical flow of information and enhances better comprehension of the system.

- **Introduction-** This chapter presents an overview of the project, including the problem statement, need, motivation, scope, and objectives. It provides a basic understanding of the proposed system and its significance.
- **Literature Survey-** This chapter reviews existing systems and research related to face recognition, monitoring, and alert systems. It highlights the limitations of current approaches and the need for the proposed solution.
- **System Design-** This chapter explains the architecture of the system along with its modules and overall design. It includes diagrams and detailed descriptions of system components and workflow.
- **Implementation-** This chapter describes the technologies used and the step-by-step development process of the system. It explains how the proposed system is built and integrated using various tools.
- **Results and Discussion-** This chapter presents the output of the system along with performance analysis. It discusses the effectiveness, accuracy, and advantages of the implemented system.
- **Conclusion and Future Work-** This chapter summarizes the overall project and its outcomes. It also suggests possible enhancements and future improvements for the system.

2. LITERATURE SURVEY

2.1 INTRODUCTION

The literature survey is an essential part of any research or project work, as it provides a comprehensive understanding of existing technologies, methodologies, and systems related to the proposed work. It helps in identifying research gaps, understanding current limitations, and justifying the need for a new or improved system. In the context of the Smart Classroom Monitoring System, the literature survey focuses on three major domains: face recognition-based systems, smart classroom monitoring systems, and automated alert mechanisms.

With the rapid advancement of technologies such as artificial intelligence, machine learning, and computer vision, several intelligent systems have been developed to automate tasks in educational institutions. These technologies have significantly improved the efficiency and accuracy of processes such as attendance management, surveillance, and communication. Face recognition, in particular, has emerged as a powerful tool for identifying individuals based on their facial features, making it highly suitable for applications like attendance tracking and identity verification.

In recent years, many educational institutions have started adopting automated attendance systems to reduce manual effort and improve accuracy. These systems use cameras and image processing techniques to detect and recognize faces, thereby marking attendance automatically. While these solutions address the problem of manual attendance, they are limited in scope as they do not provide continuous monitoring of classroom activities or ensure the presence of faculty members throughout the session.

Similarly, smart classroom monitoring systems have been developed to provide real-time supervision of classrooms. These systems often use CCTV cameras, sensors, and IoT devices to monitor activities within the classroom environment. Administrators can remotely access live feeds and ensure that classrooms are functioning properly. However, such systems primarily focus on surveillance and do not incorporate intelligent mechanisms to detect the presence or absence of faculty members. Additionally, continuous video recording raises concerns related to privacy and data storage.

Another important area in this domain is automated alert systems, which are widely used for communication purposes. These systems use technologies such as SMS gateways and email services to send notifications and reminders. In educational institutions, alert systems are used to inform students and faculty about schedules, examinations, and important announcements. Although these systems are effective in improving communication, they are generally standalone and lack integration with monitoring systems. As a result, they cannot automatically trigger alerts based on real-time events such as faculty absence.

From the study of these existing systems, it is evident that most solutions focus on a single functionality, such as attendance tracking, monitoring, or notification. There is a lack of integrated systems that combine all these features into a unified platform. Moreover, many existing systems either compromise on privacy or require significant infrastructure investment, making them less practical for widespread implementation.

Therefore, there is a need for a smart and efficient system that integrates face recognition, real-time monitoring, and automated alert mechanisms while maintaining user privacy. The proposed Smart Classroom Monitoring System aims to address these challenges by providing a comprehensive solution that ensures effective classroom supervision, improves faculty punctuality, and enhances overall academic efficiency.

2.2 LITERATURE SURVEY ON FACE RECOGNITION-BASED SYSTEMS

Face recognition technology has gained significant attention in recent years due to its wide range of applications in security, authentication, and attendance management systems. In educational institutions, face recognition-based systems are primarily used to automate the process of attendance marking. These systems rely on image processing and machine learning techniques to detect and identify individuals based on their facial features.

A typical face recognition system consists of two main stages: face detection and face recognition. Face detection involves identifying the presence of a human face in an image or video frame, while face recognition involves matching the detected face with stored data to identify the individual. Various algorithms have been developed for these purposes, each with its own advantages and limitations.

One of the commonly used techniques for face detection is the Haar Cascade classifier, which is based on machine learning and is capable of detecting faces in real time. It is widely used due to its speed and efficiency. For face recognition, algorithms such as Local Binary Patterns Histogram (LBPH) are popular due to their simplicity and effectiveness in handling variations in lighting conditions. In recent years, deep learning-based models such as Convolutional Neural Networks (CNNs) have also been used to achieve higher accuracy in face recognition tasks.

These systems have significantly improved the accuracy and efficiency of attendance management by eliminating manual processes. They reduce human errors, prevent proxy attendance, and save time for both faculty and students. Additionally, they provide a reliable and automated method for maintaining attendance records.

However, despite these advantages, face recognition-based attendance systems have certain limitations. Most of these systems are designed solely for marking attendance at a specific point in time, such as the beginning of a class. They do not provide continuous monitoring to ensure that the faculty member remains present throughout the session. As a result, they cannot detect situations where a class is left unattended after attendance has been marked.

Furthermore, these systems are not integrated with other functionalities such as timetable management or alert mechanisms. They do not send notifications or reminders to faculty members regarding their scheduled classes. This lack of integration limits their effectiveness in ensuring proper classroom supervision.

Observations:

- Provides accurate and automated attendance tracking
- Reduces manual effort and minimizes errors
- Prevents proxy attendance and improves reliability
- Limited to attendance functionality only
- Does not provide real-time classroom supervision
- No integration with timetable or alert systems

2.3 LITERATURE SURVEY ON SMART MONITORING SYSTEMS

Smart monitoring systems have been widely adopted in various domains, including education, to improve supervision and management. In the context of classrooms, these systems use technologies such as CCTV cameras, sensors, and Internet of Things (IoT) devices to monitor activities in real time. They enable administrators to remotely observe classroom environments and ensure that academic activities are conducted smoothly.

These systems typically involve the use of surveillance cameras installed in classrooms, which capture live video feeds. The video data is transmitted to a central monitoring system, where it can be viewed by administrators or management staff. Some advanced systems also use artificial intelligence techniques to analyse the video data and detect specific activities, such as movement, crowd behaviour, or unusual events.

Smart monitoring systems are particularly useful in large campuses where manual supervision is not feasible. They allow administrators to monitor multiple classrooms simultaneously and respond quickly to any issues. This improves overall efficiency and helps maintain discipline within the institution.

However, despite their advantages, these systems have several limitations. One of the major concerns is privacy. Continuous video recording of classrooms can raise ethical issues and may not be acceptable in all institutions. Storing large amounts of video data also requires significant storage capacity and increases system complexity.

Another limitation is that these systems primarily focus on surveillance rather than intelligent monitoring. They do not specifically identify or verify the presence of faculty members using advanced techniques such as face recognition. As a result, they cannot automatically determine whether a classroom is being supervised by a faculty member.

Additionally, the implementation cost of such systems can be high, especially when multiple cameras and advanced infrastructure are required. This makes them less accessible for smaller institutions with limited resources.

Observations:

- Enables real-time monitoring of classrooms
- Suitable for large campuses and institutions
- Allows remote supervision by administrators
- Raises privacy concerns due to continuous recording
- Requires high infrastructure and maintenance cost
- Does not focus on faculty presence detection

2.4 LITERATURE SURVEY ON AUTOMATED ALERT SYSTEMS

Automated alert systems play a crucial role in modern applications by enabling timely communication and notifications without manual intervention. These systems are widely used across various domains such as healthcare, banking, transportation, and education to deliver important information to users in real time. In educational institutions, automated alert systems are primarily used to notify students and faculty about class schedules, examination dates, assignment deadlines, and other academic activities.

Most automated alert systems are built using communication technologies such as Short Message Service (SMS), email services, and mobile notifications. Among these, SMS-based alert systems are highly preferred due to their reliability, simplicity, and ability to function without internet connectivity. APIs such as SMS gateways are commonly integrated into web applications to send notifications automatically based on predefined conditions or schedules.

In the context of educational systems, alert mechanisms are often used to remind faculty and students about upcoming classes or important events. These systems improve punctuality, reduce the chances of missed sessions, and enhance overall communication within the institution. For example, a timetable-based alert system can send reminders to faculty members before their scheduled classes, ensuring that they arrive on time and reducing the likelihood of unattended classrooms.

Despite these advantages, most existing automated alert systems operate independently and are not integrated with real-time monitoring or detection systems. They rely on static schedules and predefined triggers rather than dynamic inputs from real-world events. As a result, they are unable to respond intelligently to situations such as faculty absence or unsupervised classrooms.

Another limitation is the lack of contextual awareness in these systems. They do not verify whether a faculty member is actually present in the classroom before sending alerts or notifications. This reduces their effectiveness in ensuring proper classroom supervision. Additionally, these systems do not provide analytical insights such as attendance trends or performance reports, which are essential for institutional decision-making.

To overcome these limitations, it is necessary to integrate automated alert systems with intelligent technologies such as face recognition and real-time monitoring. By combining these features, alerts can be triggered based on actual events rather than predefined schedules. For instance, if a faculty member is not detected in the classroom during a scheduled time, the system can automatically send an alert, making the system more efficient and responsive.

The proposed Smart Classroom Monitoring System enhances the functionality of traditional alert systems by integrating them with face recognition and timetable data. It not only sends reminders before classes but also ensures that alerts are meaningful and context-aware. This integration significantly improves the reliability and effectiveness of the alert mechanism.

Observations:

- Provides fast and reliable communication through SMS or notifications
- Improves punctuality and reduces missed classes
- Easy to implement using APIs and web technologies
- Works well for schedule-based reminders
- Lacks integration with real-time monitoring systems
- Cannot detect or respond to real-world events automatically

2.5 LITERATURE SURVEY ON ATTENDANCE ANALYSIS AND REPORTING SYSTEMS

In recent years, educational institutions have increasingly focused on not only recording attendance but also analysing it to improve faculty performance and institutional efficiency. Attendance analysis and reporting systems are designed to generate meaningful insights from collected attendance data, helping administrators make informed decisions.

Traditional attendance systems primarily store records without providing analytical features. However, modern systems incorporate reporting mechanisms that generate summaries such as daily, weekly, or monthly attendance reports. These reports help in identifying patterns such as faculty punctuality, absenteeism, and consistency in conducting classes.

Many existing systems use database management techniques to store attendance data and generate reports in the form of tables, charts, or downloadable files. Some advanced systems also use data analytics techniques to visualize trends and provide insights into performance metrics. These features are particularly useful for academic administrators to monitor faculty behaviour and ensure accountability.

Despite these advancements, most attendance reporting systems are limited in their integration with real-time monitoring technologies. They often rely on manually entered data or basic attendance logs, which may not always be accurate. Additionally, these systems do not combine attendance analysis with intelligent detection methods such as face recognition.

Another limitation is that many systems provide only basic reporting features and lack detailed analysis over shorter time periods such as weekly performance tracking. Weekly reports are important as they allow institutions to quickly identify irregularities and take corrective actions without waiting for long-term summaries.

The proposed Smart Classroom Monitoring System addresses these limitations by integrating attendance tracking with an automated reporting mechanism. The system generates weekly faculty attendance reports based on real-time face recognition data. This ensures that the reports are accurate and reflect actual classroom presence.

The weekly report feature enables administrators to:

- Monitor faculty punctuality and consistency
- Identify attendance patterns and irregularities
- Evaluate faculty performance over time
- Make data-driven decisions for academic improvement

By combining real-time monitoring with analytical reporting, the system provides a more comprehensive solution compared to existing systems. This feature enhances accountability and supports better management of academic activities.

Observations:

- Provides structured analysis of attendance data
- Helps in tracking faculty performance and consistency
- Useful for administrative decision-making
- Improves accountability and transparency
- Most existing systems rely on manual or static data
- Lack integration with real-time detection systems
- Limited focus on short-term (weekly) performance analysis

2.6 LITERATURE SURVEY ON ROLE-BASED ACCESS CONTROL

Role-Based Access Control (RBAC) is a widely used security mechanism in modern software systems, especially in web applications and enterprise systems. It restricts system access based on predefined roles assigned to users, ensuring that only authorized individuals can access specific resources or perform certain operations.

According to research, RBAC is designed such that permissions are assigned to roles, and users are assigned to those roles, rather than assigning permissions directly to individual users. This approach simplifies security management and reduces administrative complexity in large systems.

Early studies by NIST highlighted that RBAC significantly reduces the complexity of managing access control compared to traditional methods like Access Control Lists (ACLs), which require individual permission assignment for each user. RBAC is particularly effective in web-based systems where multiple users interact with shared resources.

Modern research shows that RBAC follows key principles such as:

- Authentication – verifying user identity
- Authorization – granting access based on roles
- Role assignment – assigning users to roles
- Permission management – defining what each role can access

In web applications, RBAC is commonly implemented using frameworks like Flask or Django, where different roles such as Admin, Faculty, and User are defined. Each role has specific permissions, such as viewing reports, managing data, or accessing dashboards.

Observations:

- RBAC simplifies access control management in large systems
- Enhances security by restricting unauthorized access
- Supports scalability and flexibility in web applications
- Reduces administrative overhead compared to traditional methods
- Ensures implementation of least privilege principle
- May become complex if too many roles are defined

2.7 COMPARATIVE ANALYSIS OF EXISTING SYSTEMS

Parameters	Face Recognition Attendance Systems	Smart Monitoring Systems	Automated Alert Systems	Attendance Reporting Systems	Proposed System
Primary Function	Attendance marking using face recognition	Classroom surveillance and monitoring	Sending reminders/notifications	Attendance analysis and reporting	Integrated monitoring + alerts + reporting
Technology Used	Image processing, ML algorithms	CCTV, IoT, sensors	SMS/Email APIs	Databases, analytics tools	AI, OpenCV, Flask, JSON, SMS API
Real-Time Monitoring	Not available	Available	Not available	Not available	Available
Faculty Presence Detection	Yes (at entry only)	No	No	No	Yes (continuous detection)
Automated Alerts	No	No	Yes	No	Yes (timetable-based SMS alerts)
Timetable Integration	No	No	Limited	No	Yes
Privacy Friendly	Yes	No (continuous video recording)	Yes	Yes	Yes (no video storage, only detection)
Cost Efficiency	Moderate	High	Low	Low	Moderate

Parameters	Face Recognition Attendance Systems	Smart Monitoring Systems	Automated Alert Systems	Attendance Reporting Systems	Proposed System
Analytical Reports	No	No	No	Yes	Yes (daily/weekly reports)
System Integration	Limited	Limited	Standalone	Standalone	Fully Integrated
Role-Based Login (Secure Access)	Not available	Limited (basic access control)	Not available	Limited	Yes (Admin & Faculty roles)
Ease of Implementation	Medium	Complex	Easy	Easy	Medium
Scalability	Medium	High	High	High	Medium (can be extended with database)

Table-2.1: Comparative Analysis of Existing Systems

2.8 CONCLUSION

The literature survey shows that existing systems focus on separate functionalities like attendance, monitoring, alerts, and reporting, without a unified and secure approach. While face recognition improves accuracy and monitoring enables supervision, issues like lack of integration and privacy concerns remain. Additionally, most systems do not include proper access control mechanisms.

The proposed system addresses these gaps by integrating face recognition, real-time monitoring, SMS alerts, reporting, and Role-Based Access Control (RBAC). RBAC ensures secure access by defining roles such as Admin and Faculty, improving data protection and system reliability.

3. SYSTEM ANALYSIS

3.1 OVERVIEW

System Analysis and Design is a fundamental phase in the software development lifecycle that focuses on understanding user requirements, analysing current systems, and designing improved solutions. It ensures that the final system meets functional and non-functional requirements efficiently while maintaining reliability, scalability, and security.

In the context of the Smart Classroom Monitoring System, system analysis involves studying traditional classroom management techniques and identifying inefficiencies such as manual attendance, lack of monitoring, and delayed communication. The design phase then proposes an intelligent system that integrates modern technologies like Artificial Intelligence (AI), Machine Learning (ML), and computer vision.

The proposed system is designed to automate and enhance classroom operations by incorporating face recognition-based attendance, real-time monitoring, alert mechanisms, and report generation. Additionally, the integration of Role-Based Access Control (RBAC) ensures that data security and privacy are maintained by restricting access to authorized users only.

This chapter provides a comprehensive understanding of the existing system, its limitations, the proposed solution, and feasibility analysis to justify the implementation of the system.

3.2 EXISTING SYSTEM

The existing classroom monitoring systems are mostly based on manual observation or basic CCTV surveillance. Teachers or administrators monitor classrooms either physically or by reviewing recorded video footage.

These systems are not automated and lack intelligence. Although CCTV cameras record activities, they do not provide real-time alerts or meaningful analysis. Human intervention is required to monitor and analyse the recordings, which makes the process inefficient.

3.2.1 CHALLENGES IN EXISTING SYSTEM

The existing system faces several limitations:

- It requires continuous human monitoring.
- It does not provide real-time alerts for unusual activities.
- Reviewing CCTV footage is time-consuming.
- Human errors can affect monitoring accuracy.
- No automation in attendance tracking.
- Lack of intelligent data analysis.
- Difficult to manage multiple classrooms efficiently.
- Security risks due to delayed response.

These challenges indicate the need for a more efficient and automated system.

3.3 PROPOSED SYSTEM

The proposed Smart Classroom Monitoring System is an advanced solution that integrates artificial intelligence and computer vision technologies. It is designed to automate classroom monitoring and improve efficiency.

The system uses cameras to capture live video from classrooms. This video is processed using machine learning algorithms to identify students through face recognition and monitor their activities.

The system can automatically mark attendance, detect unusual behaviour, and send alerts to authorities. It also stores data for future analysis and provides a dashboard for easy monitoring.

This system reduces manual effort and ensures accurate and real-time monitoring.

3.3.1 SYSTEM OBJECTIVES

The primary objectives of the Smart Classroom Monitoring System are as follows:

- To provide continuous real-time monitoring of classroom activities.
- To ensure immediate communication through automated alerts and notifications.
- To enhance data security using Role-Based Access Control (RBAC).
- To generate accurate and comprehensive reports for analysis.
- To reduce manual workload and improve productivity of faculty and administrative staff.
- To improve overall classroom management and operational efficiency.
- To provide a scalable and flexible system that can be implemented across various institutions.

3.3.2 FEATURES OF PROPOSED SYSTEM

The proposed system includes the following features:

- Face Recognition: Automatically identifies students and marks attendance.
- Real-time Monitoring: Tracks classroom activities continuously.
- Motion Detection: Identifies unusual movements or disturbances.
- Automated Alerts: Sends notifications during abnormal situations.
- Data Storage: Maintains records for future reference.
- User Interface: Provides an easy-to-use dashboard for monitoring.
- Scalability: Can be extended to multiple classrooms.
- High Accuracy: Reduces errors through automation.
- Role-Based Access: Restricts access based on roles:
 - a) admin-full access
 - b) faculty- only reports

3.4 FEASIBILITY STUDY

The feasibility study evaluates whether the proposed system is practical and beneficial. It is divided into three categories: technical, operational, and economic feasibility.

3.4.1 TECHNICAL FEASIBILITY

- The system uses technologies such as Python, machine learning, and computer vision.
- Libraries like OpenCV and TensorFlow support face recognition and image processing.
- These tools are open-source and easy to implement.
- The required hardware (cameras and computer systems) is readily available.
- The system can be integrated with existing infrastructure without major changes.
- Overall, the system is technically feasible and reliable.

3.4.2 OPERATIONAL FEASIBILITY

- The system is user-friendly and easy to operate.
- Teachers and administrators require minimal technical knowledge.
- Very little training is needed to use the system.
- Automation reduces manual effort and workload.
- Real-time alerts help in quick decision-making.
- The system improves classroom discipline and monitoring efficiency.

3.4.3 ECONOMIC FEASIBILITY

- The system requires initial investment for hardware setup.
- Open-source software reduces development costs.
- Automation reduces labour costs for monitoring.
- It saves time in attendance and surveillance tasks.
- Maintenance costs are relatively low.
- The system is cost-effective in the long run.

3.5 CONCLUSION

This chapter presented a detailed analysis and design of the Smart Classroom Monitoring System, highlighting the limitations of existing systems and the need for an automated and intelligent solution. The proposed system integrates advanced technologies such as artificial intelligence, computer vision, and Role-Based Access Control (RBAC) to enhance monitoring and ensure secure access.

The system incorporates features like face recognition, real-time monitoring, automated alerts, and report generation, which improve accuracy, efficiency, and overall system reliability. It reduces manual effort and streamlines classroom management processes.

The feasibility study confirms that the system is technically achievable, operationally convenient, and economically viable. Overall, the proposed system provides a modern, scalable, and secure solution, supporting efficient monitoring and better decision-making through data analysis.

Overall, the proposed system provides a modern, reliable, and scalable solution for educational institutions. It not only improves monitoring but also supports better decision-making through data analysis. Hence, the Smart Classroom Monitoring System is an effective step towards building smart and secure learning environments.

4. DESIGN PROCESS

4.1 SYSTEM ARCHITECTURE

The Smart Classroom Monitoring System is designed using a modular and scalable architecture that integrates multiple technologies such as face recognition, real-time video processing, alert systems, and data analytics. The system follows a layered architecture consisting of presentation layer, application layer, and data layer.

The Presentation Layer provides the user interface through which administrators and faculty interact with the system. It includes dashboards, attendance reports, and alert notifications.

The Application Layer handles core functionalities such as face detection, recognition, attendance processing, and alert generation. It acts as a bridge between user inputs and database operations.

The Data Layer is responsible for storing and managing student records, attendance logs, and system-generated reports. It ensures secure and efficient data retrieval.

The architecture is designed to support real-time monitoring while maintaining accuracy, scalability, and security.

4.2 UML DIAGRAMS

Unified Modeling Language (UML) diagrams are used to represent the system design visually. These diagrams help in understanding system structure, behavior, and interactions.

- Class Diagram
- Use Case Diagram
- Sequence Diagram
- Deployment Diagram

4.2.1 CLASS DIAGRAM

The class diagram represents the static structure of the system by showing classes, attributes, and relationships.

Key classes include: User (Admin, Faculty), Attendance Face Recognition, Module Alert System, Database Manager.

Each class contains attributes such as ID, name, and role, and methods such as login(), markAttendance(), generateReport().

Relationships include:

- Inheritance between User, Admin, and Faculty
- Dependency between Face Recognition and Attendance modules

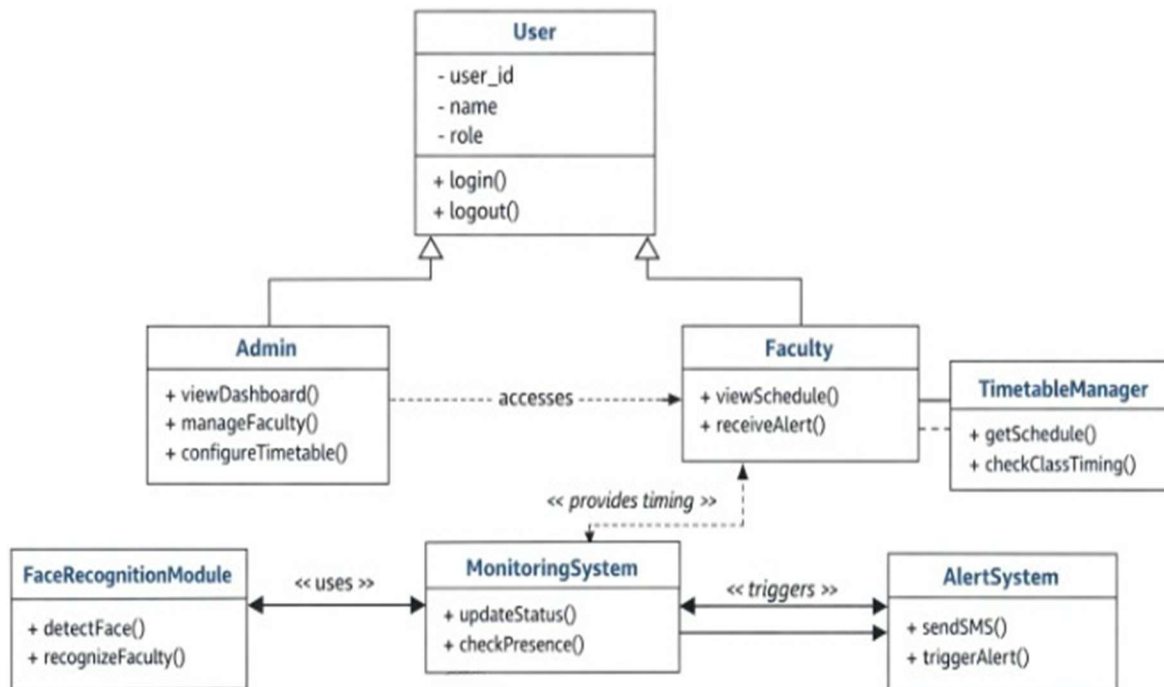


Fig-4.2.1: UML Class Diagram

4.2.2 USE CASE DIAGRAM

The use case diagram illustrates interactions between users and the system.

Main actors:

- Admin
- Faculty

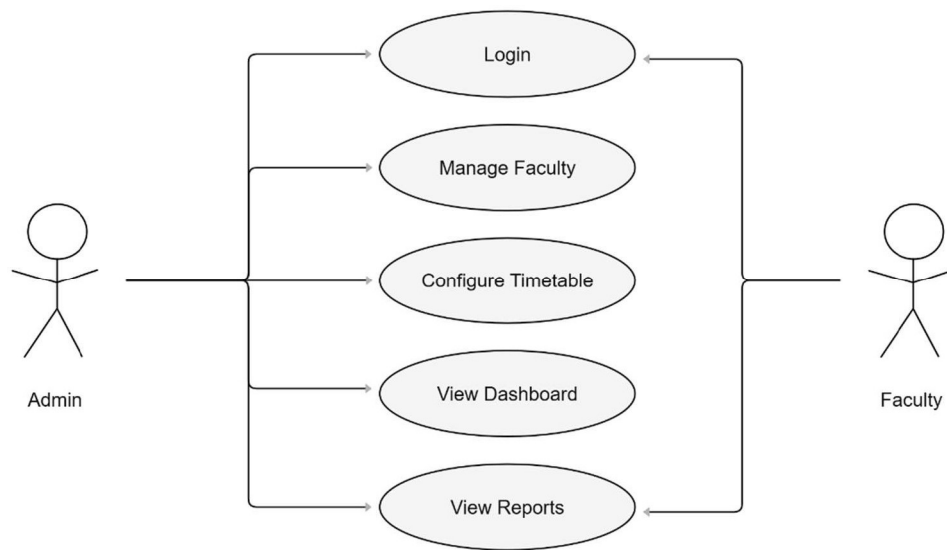


Fig-4.2.2: UML Use Case Diagram

4.2.3 SEQUENCE DIAGRAM

The sequence diagram shows the flow of operations over time.

1. User logs into system
2. Camera captures image
3. Face recognition module processes image
4. System verifies identity

5. Attendance is marked
6. Data is stored in database
7. Alerts are generated if anomalies are detected

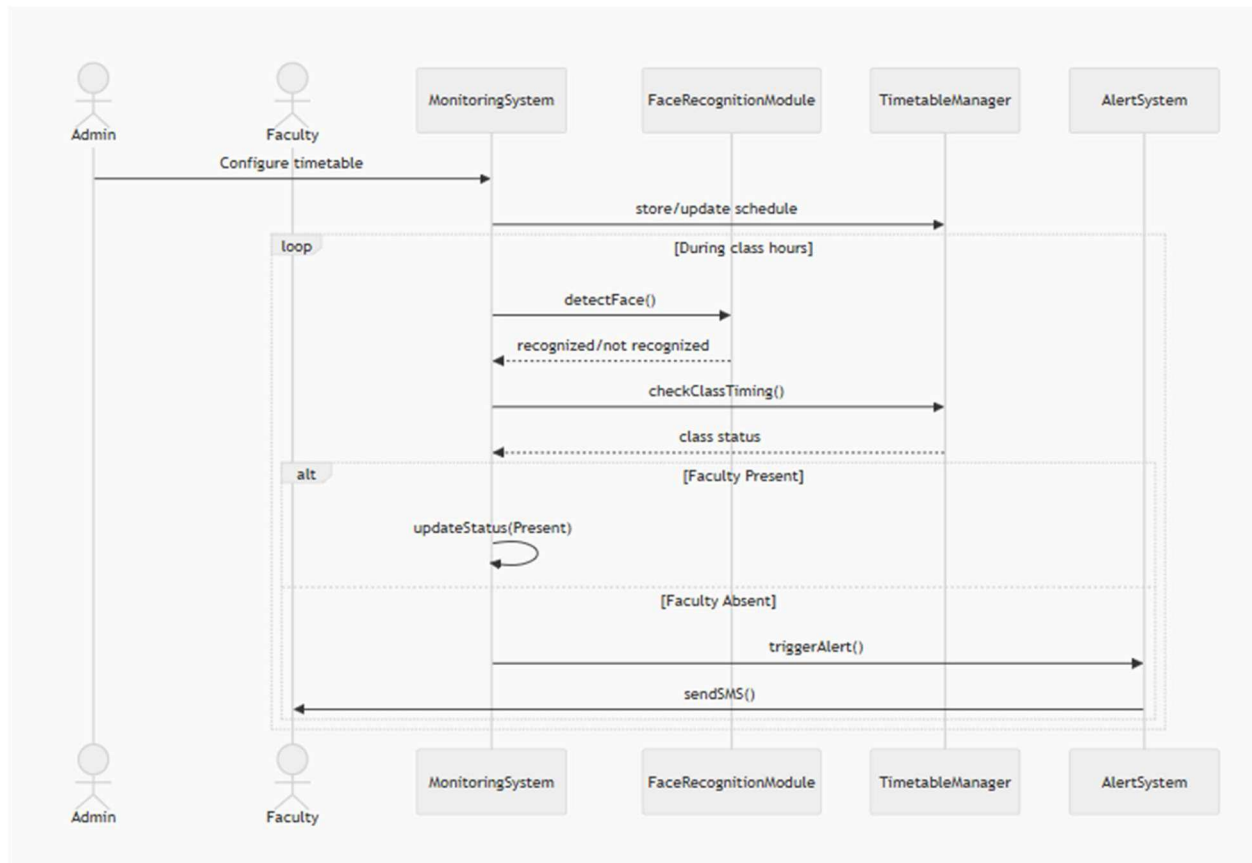


Fig-4.2.3: UML Sequence Diagram

4.2.4 DEPLOYMENT DIAGRAM

The deployment diagram represents the physical setup of the system.

Components include: Client device (web interface), Server (application processing), Camera module (input device), Database server.

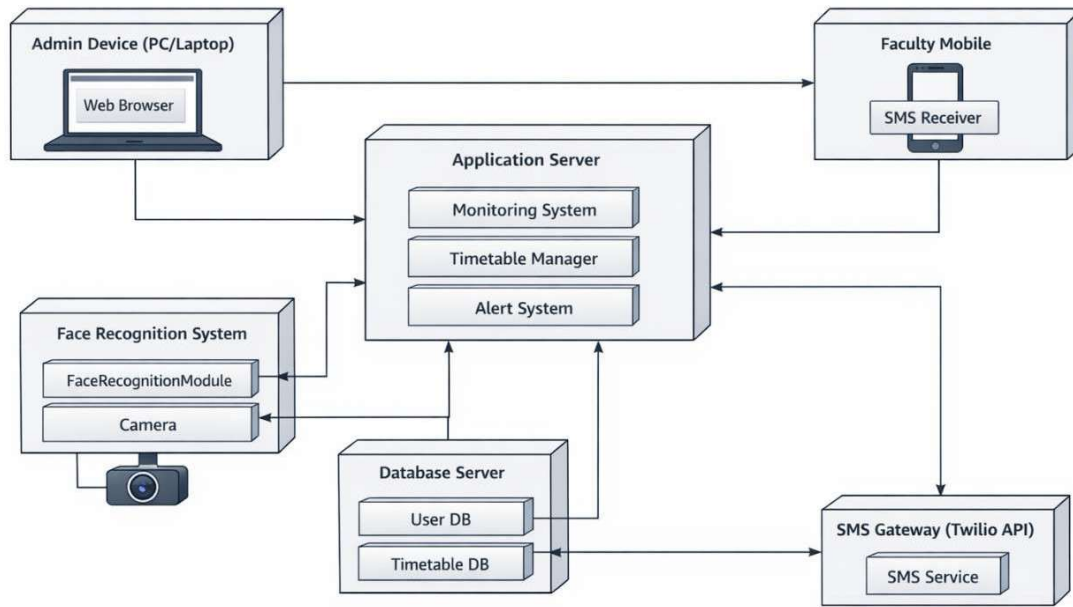


Fig-4.2.4: Deployment Diagram

4.3 ARCHITECTURE DIAGRAM

This diagram represents how modules interact within the system.

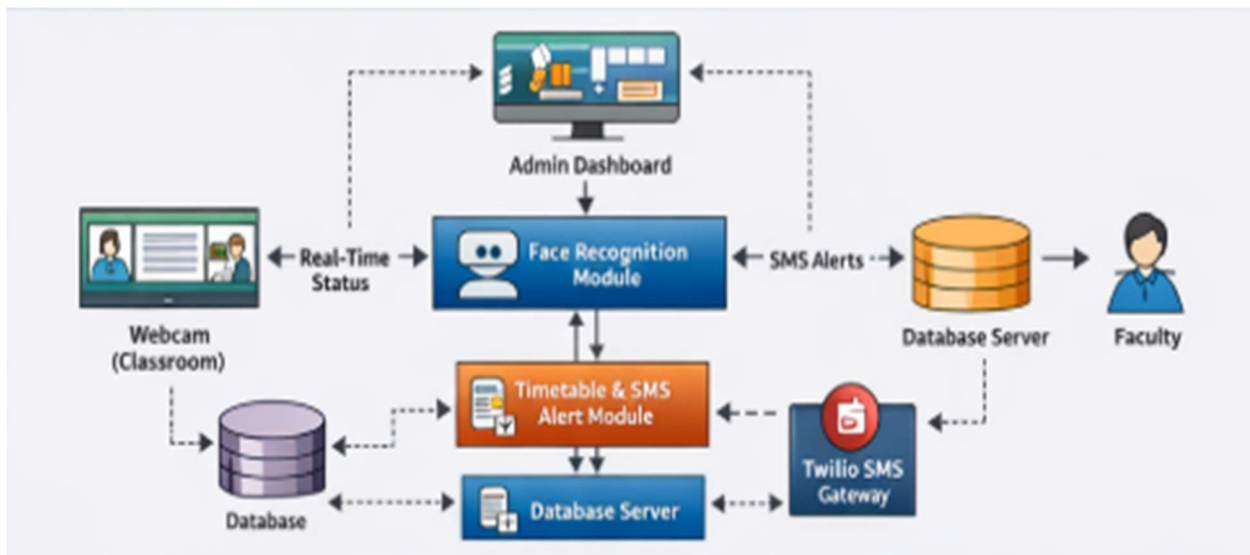


Fig-4.3: Architecture Diagram

4.3.1 MODULES DESCRIPTION

The system is divided into the following modules:

1. User Management Module (RBAC)

- Handles authentication and authorization
- Provides role-based access:
 - Admin → Full control
 - Faculty → Limited access (punctuality stats)

2. Face Recognition Module

- Captures live frames from webcam
- Detects and recognizes faculty faces
- Uses stored facial encodings for matching

3. Classroom Monitoring Module

- Tracks faculty presence in classrooms
- Updates classroom status:
 - Green → Faculty present
 - Red → Faculty absent

4. Timetable & Scheduling Module

- Stores faculty schedules
- Checks whether a class is currently active
- Triggers alerts based on schedule

5. Alert Module (SMS Integration)

- Sends SMS reminders before class
- Notifies faculty for missed or delayed sessions

6. Reporting & Analytics Module

- Tracks punctuality of faculty
- Calculates attendance percentage based on presence timing
- Accessible via RBAC

4.3.2 DATA PROCESSING & STORAGE

Processing Flow:

1. Capture image via webcam
2. Detect face using OpenCV
3. Extract facial features
4. Match with stored faculty dataset
5. Validate against timetable
6. Update classroom status

Stored Data:

- Faculty details
- Face encodings
- Timetable schedules
- Presence logs (timestamp-based)
- Alert records

4.3.3 VISUALIZATION AND ANALYTICAL MODULE

The dashboard provides:

- Real-time classroom status (green/red indicators)
- Faculty punctuality percentage
- Daily/weekly reports
- Alert logs

Admins can analyse:

- Missed classes
- Late arrivals
- Overall supervision efficiency

Faculty can view:

- Attendance percentage based on punctuality
- Bar Graphs based on percentage

4.4 Data Flow Diagram (DFD)

Level 0 (Context Diagram):

- Admin interacts with system
- Faculty receives alerts
- System processes camera input

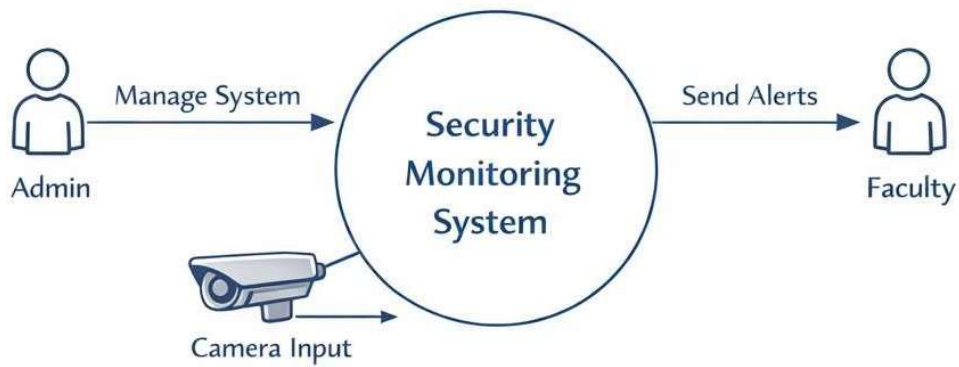


Fig-4.4.1: Level-0 Data Flow Diagram

Level 1:

- Face recognition process
- Timetable validation
- Alert generation
- Dashboard update

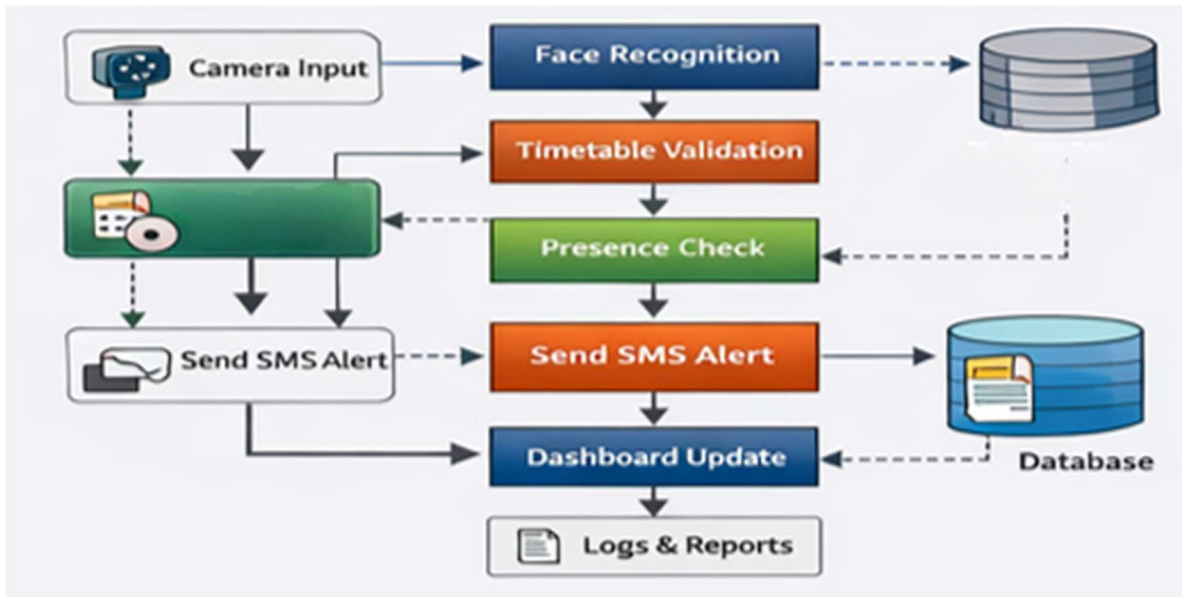


Fig-4.4.2: Level-1 Data Flow Diagram

4.5 ARCHITECTURE FLOW

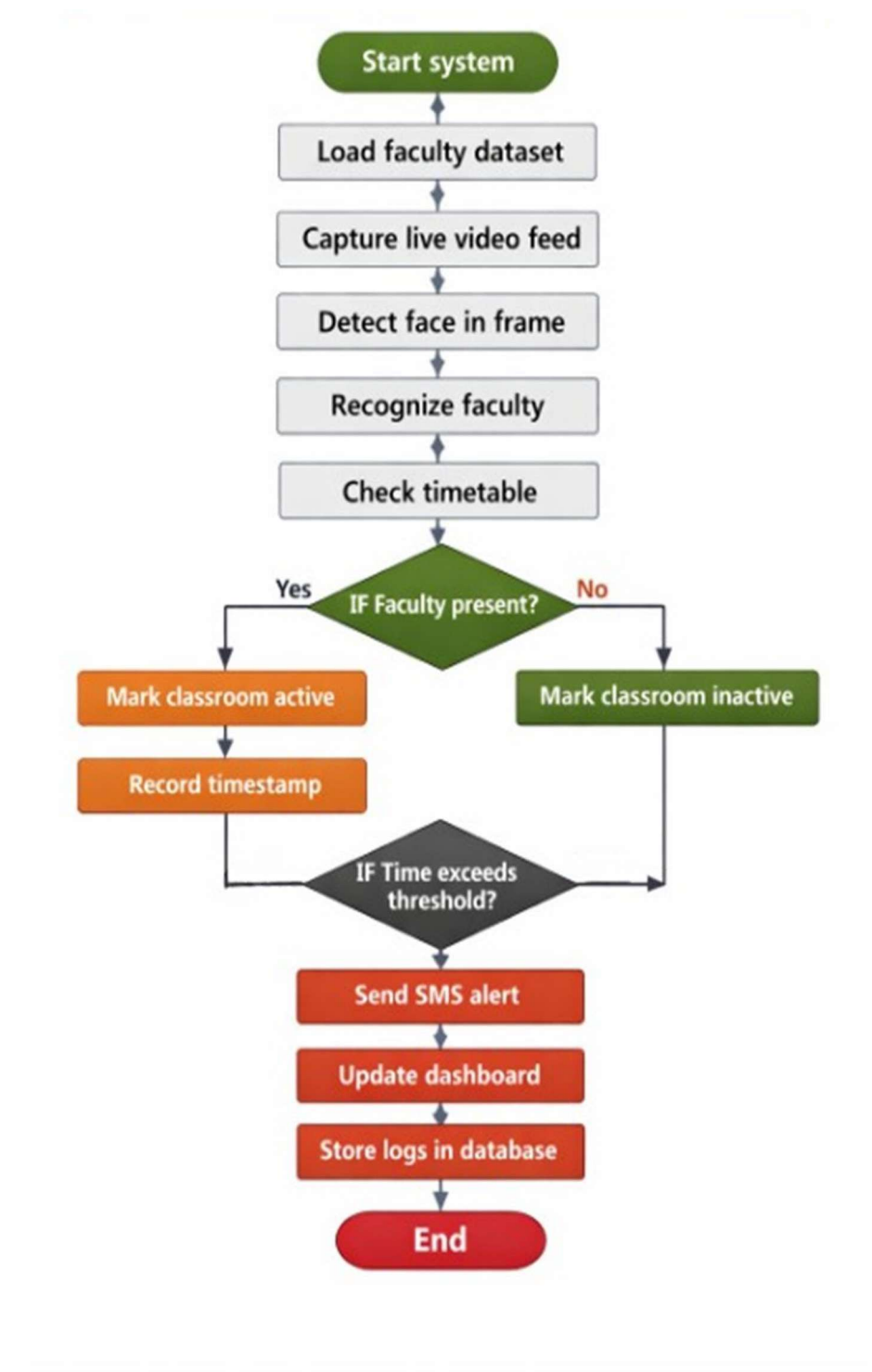


Fig-4.5: Architecture Flow

4.6 User Interface Design

The UI is designed for clarity and ease of use.

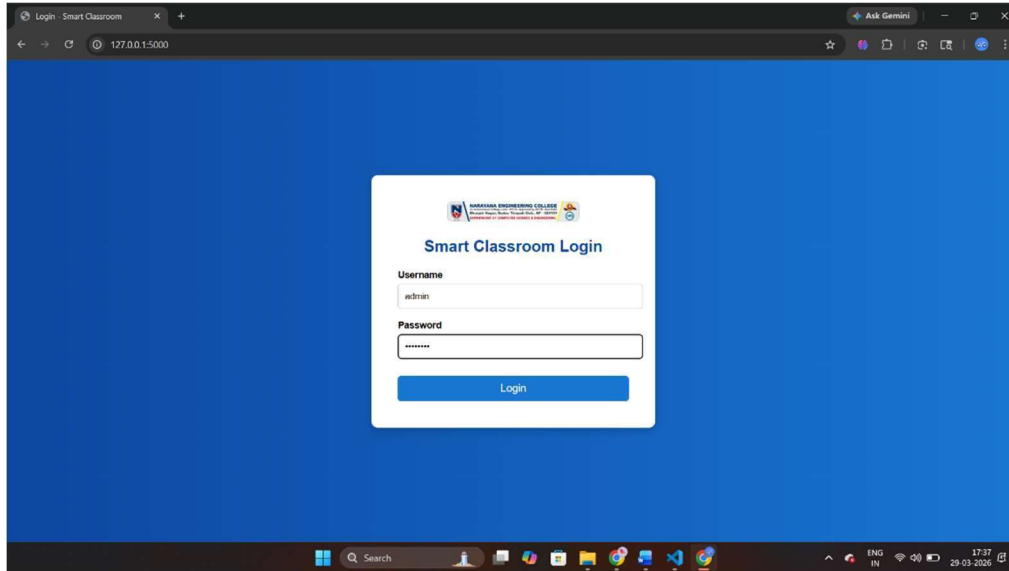


Fig-4.6.1: Login Page

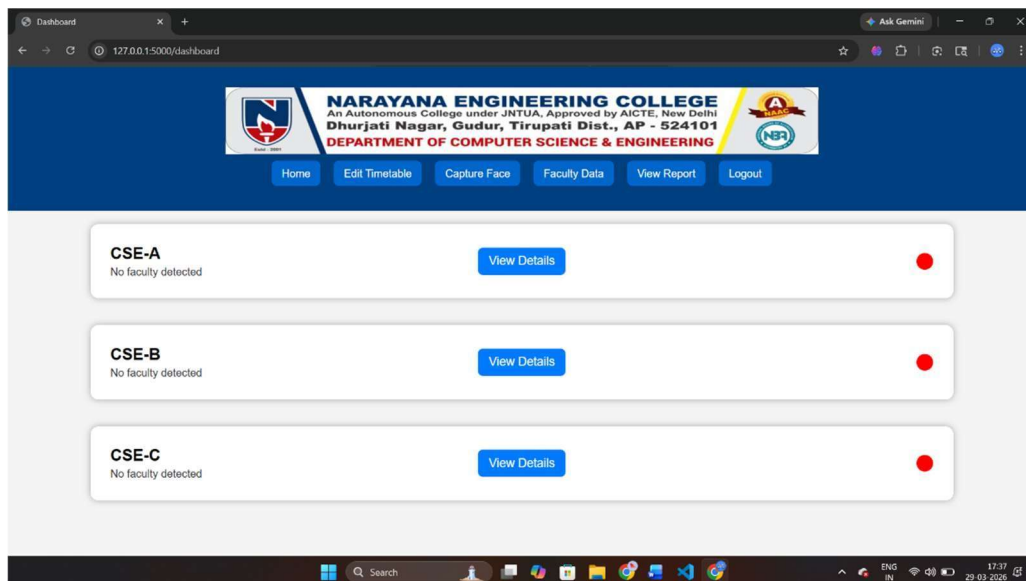


Fig-4.6.2: Main Screen

4.7 SYSTEM REQUIREMENTS

The successful implementation of the Smart Classroom Monitoring System depends on specific hardware and software requirements. These requirements ensure that the system operates efficiently, processes data accurately, and supports real-time functionalities such as face recognition and monitoring.

4.7.1 HARDWARE REQUIREMENTS

- Processor: Intel i5 or higher
- RAM: Minimum 8 GB
- Storage: 256 GB
- Webcam: HD camera for face detection
- Internet connectivity

4.7.2 SOFTWARE REQUIREMENTS

- Operating System: Windows/Linux
- Programming Language: Python
- Libraries: OpenCV, face-recognition
- Framework: Flask
- Database: MySQL / SQLite
- SMS API: Twilio

5. METHODOLOGY

5.1 INTRODUCTION

The methodology of the Smart Classroom Monitoring System defines the systematic approach used to design, develop, and implement the system. This chapter explains how various components such as face recognition, real-time monitoring, SMS alerts, and report generation are integrated into a unified platform.

The primary goal of the methodology is to ensure efficient classroom supervision, improve faculty accountability, and provide automated reporting. The system follows a modular and scalable design, allowing easy integration of new features in the future.

The development approach includes:

- Requirement analysis
- System design
- Implementation of modules
- Testing and validation

A combination of software engineering principles and modern technologies such as Artificial Intelligence (AI), database management, and web technologies are used to build the system.

5.2 TECHNOLOGY DESCRIPTION

The Smart Classroom Monitoring System is developed using a combination of frontend, backend, AI-based processing, and communication technologies.

5.2.1 FRONTEND TECHNOLOGIES

The frontend provides the user interface for Admin and Faculty.

- **HTML, CSS, JavaScript** – Used for designing responsive and interactive web pages

- **Bootstrap** – Ensures mobile-friendly and responsive design
- Provides dashboards, reports, and configuration interfaces

5.2.2 BACKEND TECHNOLOGIES

The backend handles business logic, authentication, and system operations.

- **Python**
- Handles:
 - User authentication
 - Role-Based Access Control (RBAC)
 - Data processing
 - API communication

5.2.3 DATA STORAGE

- **JSON Files**
- Stores:
 - Faculty details
 - Timetable data
 - Attendance logs
 - Monitoring records
- Ensures data consistency and secure storage

5.3 SYSTEM ARCHITECTURE

The system follows a layered architecture:

- Presentation Layer (Frontend)
- Application Layer (Backend)
- Data Layer (Data Storage)
- AI Processing Layer (Face Recognition Module)

Data flows from cameras → processing → database → alerts → reports.

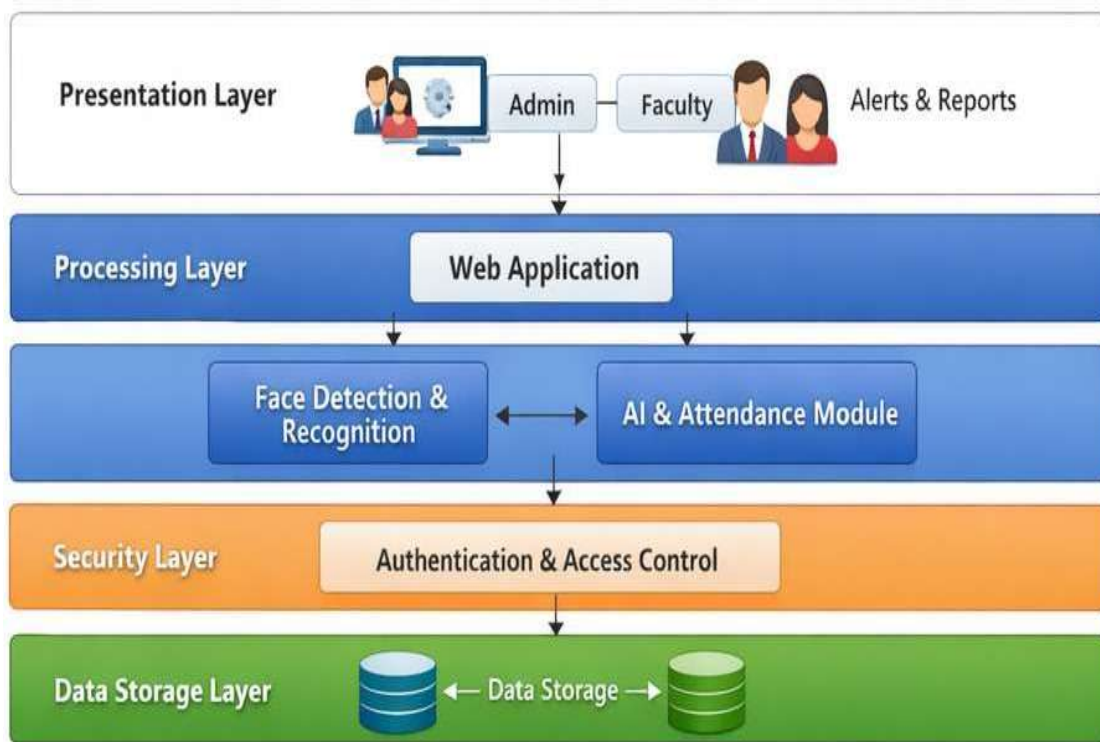


Fig-5.3: System Architecture

5.4 METHODOLOGY WORKFLOW

The working process of the system is as follows:

1. Camera captures classroom images
2. Face recognition detects faculty presence
3. Data is processed and stored in database
4. If absence is detected:
 - SMS alert is sent
5. Weekly reports are generated automatically
6. Admin monitors system through dashboard

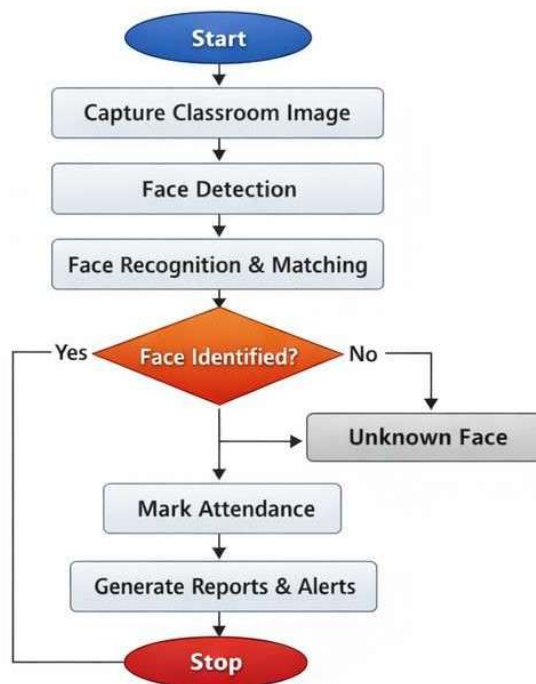


Fig-5.4: Methodology Work Flow Diagram

5.5 ADVANTAGES OF METHODOLOGY

- Real-time monitoring
- Automated alerts and reporting
- Reduced manual effort
- Improved accountability
- Scalable and modular design

5.6 LIMITATIONS

- Dependence on camera quality
- Lighting conditions affect accuracy
- Internet required for alerts
- Privacy concerns (handled with controlled access)

6. RESULTS AND DISCUSSIONS

6.1 INTRODUCTION

This chapter presents the outcomes obtained after the implementation of the Smart Classroom Monitoring System. It evaluates the system's performance based on various parameters such as accuracy, efficiency, usability, and reliability. The discussion also highlights how effectively the system meets the objectives defined in earlier chapters, including real-time monitoring, automated alerts, and report generation.

6.2 SYSTEM IMPLEMENTATION RESULTS

The system was successfully developed and deployed using the proposed architecture. The major modules such as face recognition, classroom monitoring, SMS alerts, and report generation were tested under real-time conditions.

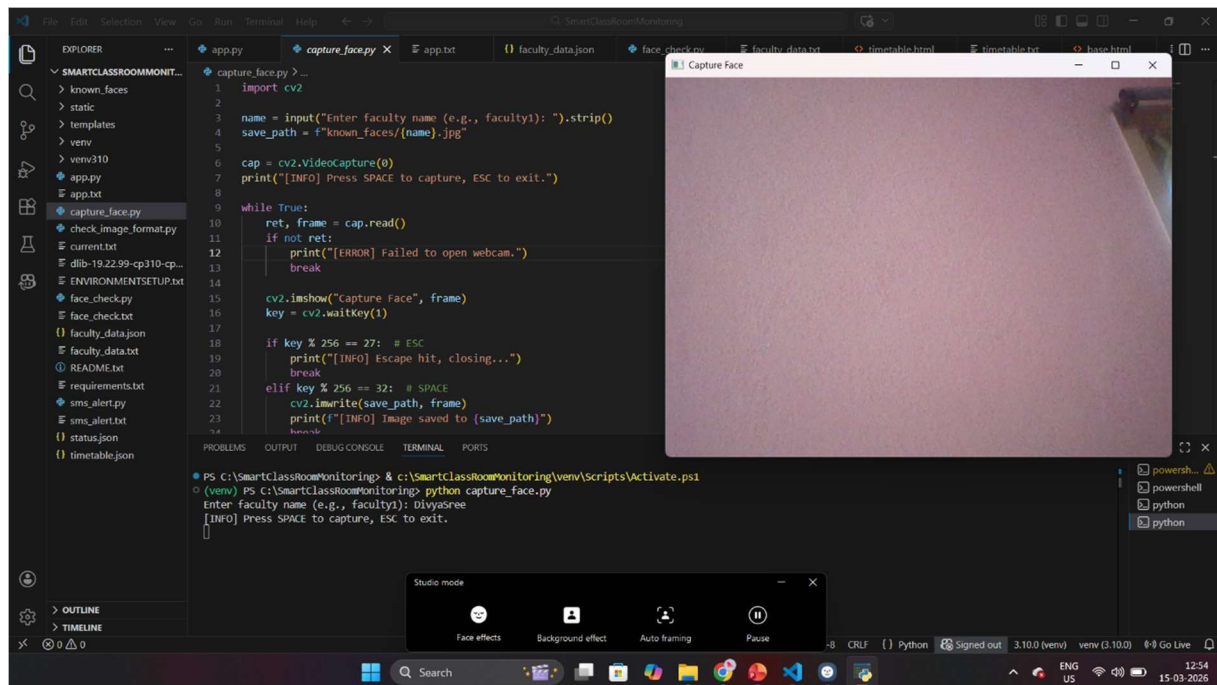


Fig-6.2.1: Face Capture Screenshot

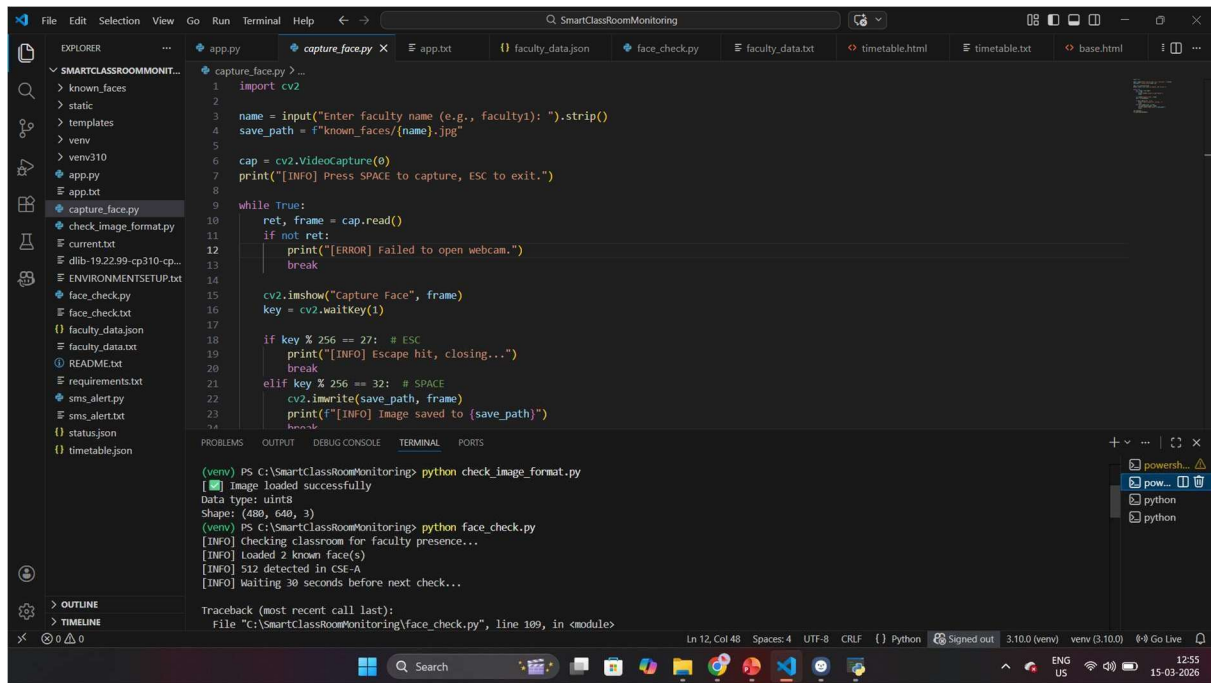


Fig-6.2.2: Checking Image Format

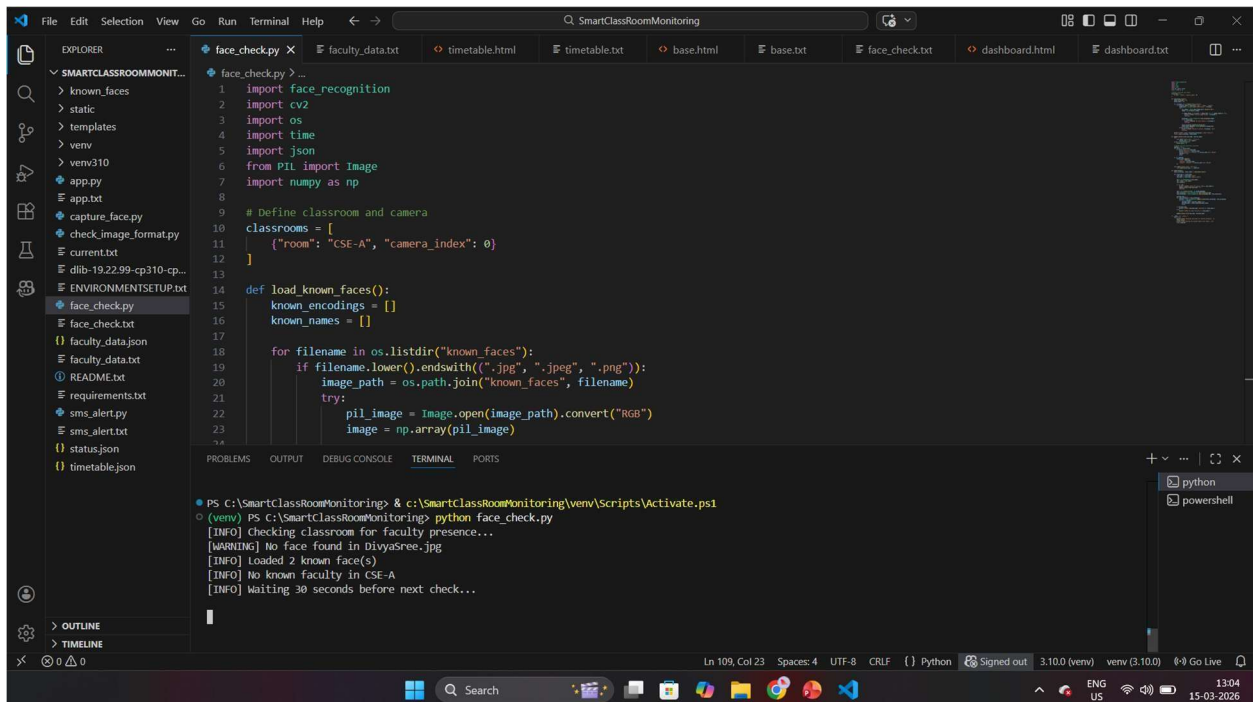


Fig-6.2.3: Face Check

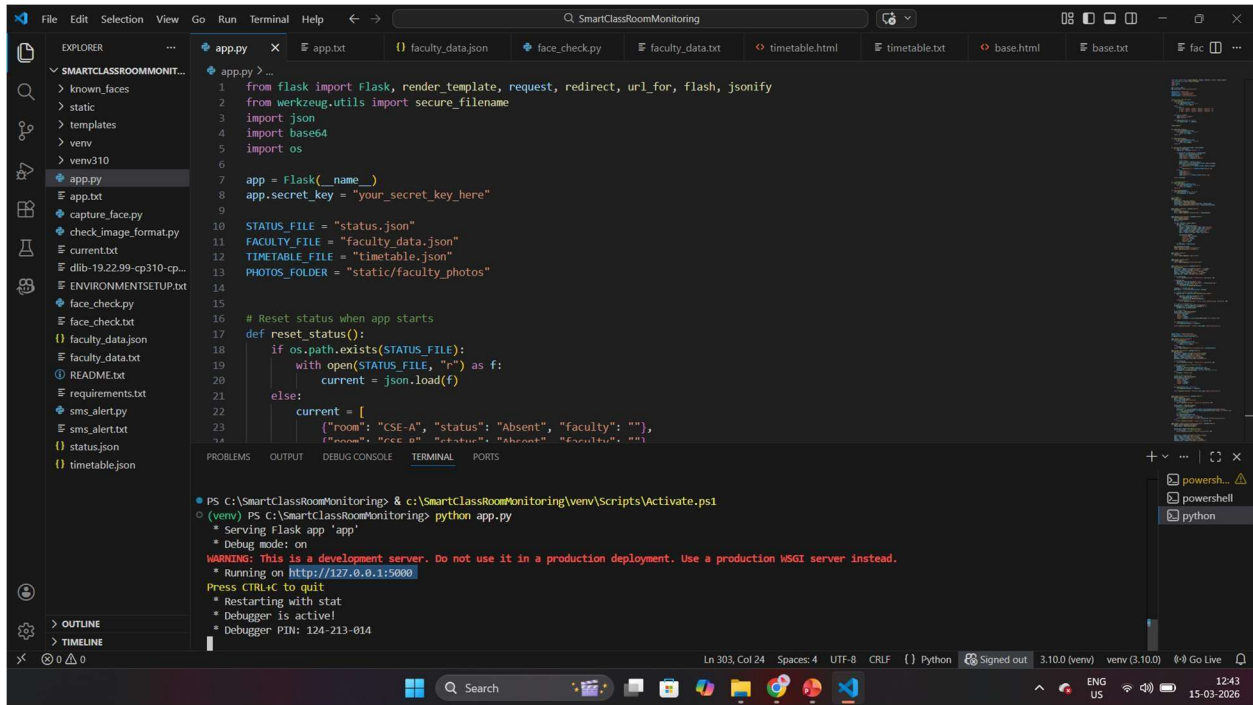


Fig-6.2.4: app.py Capture

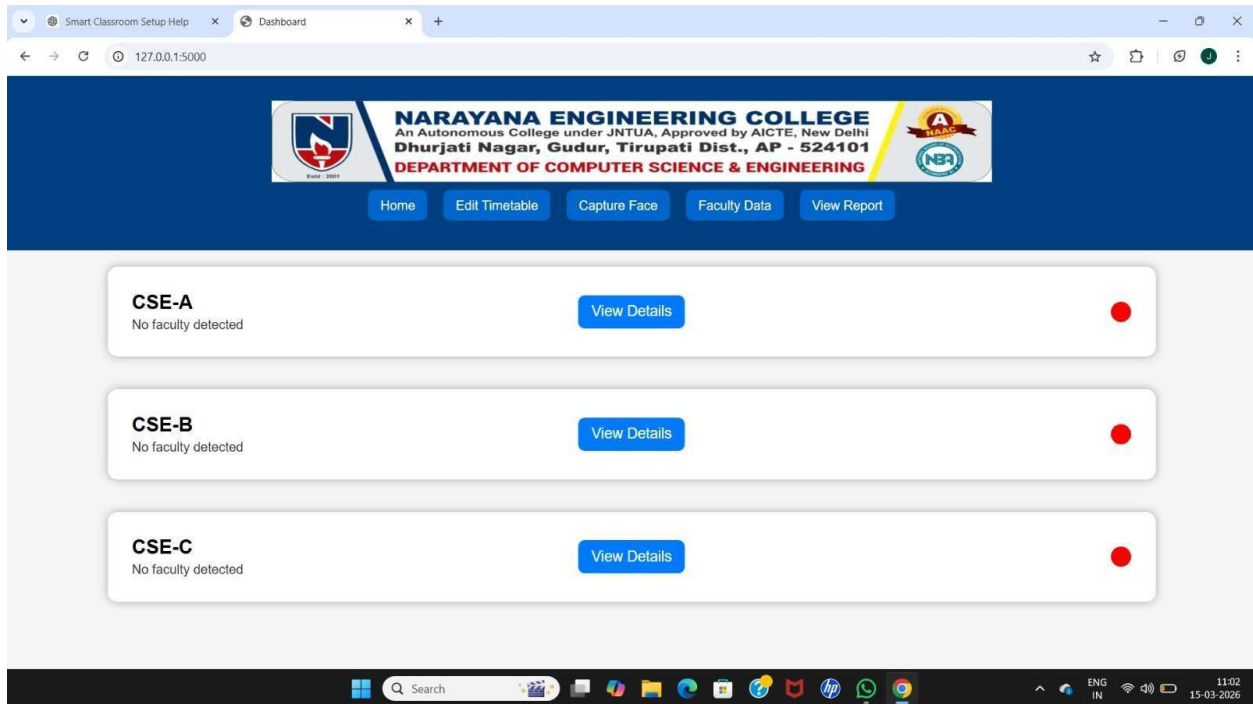


Fig-6.2.5: Home Page (Before Detection)

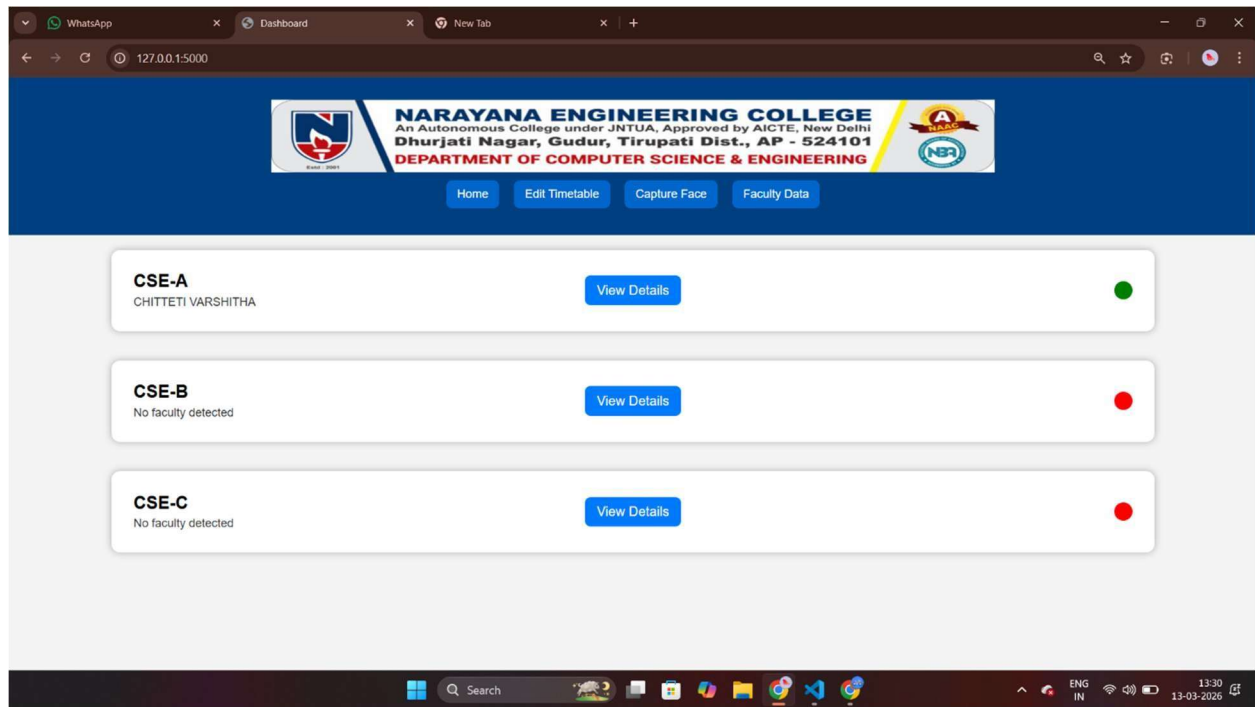


Fig-6.2.6: Home Page (After Detection)

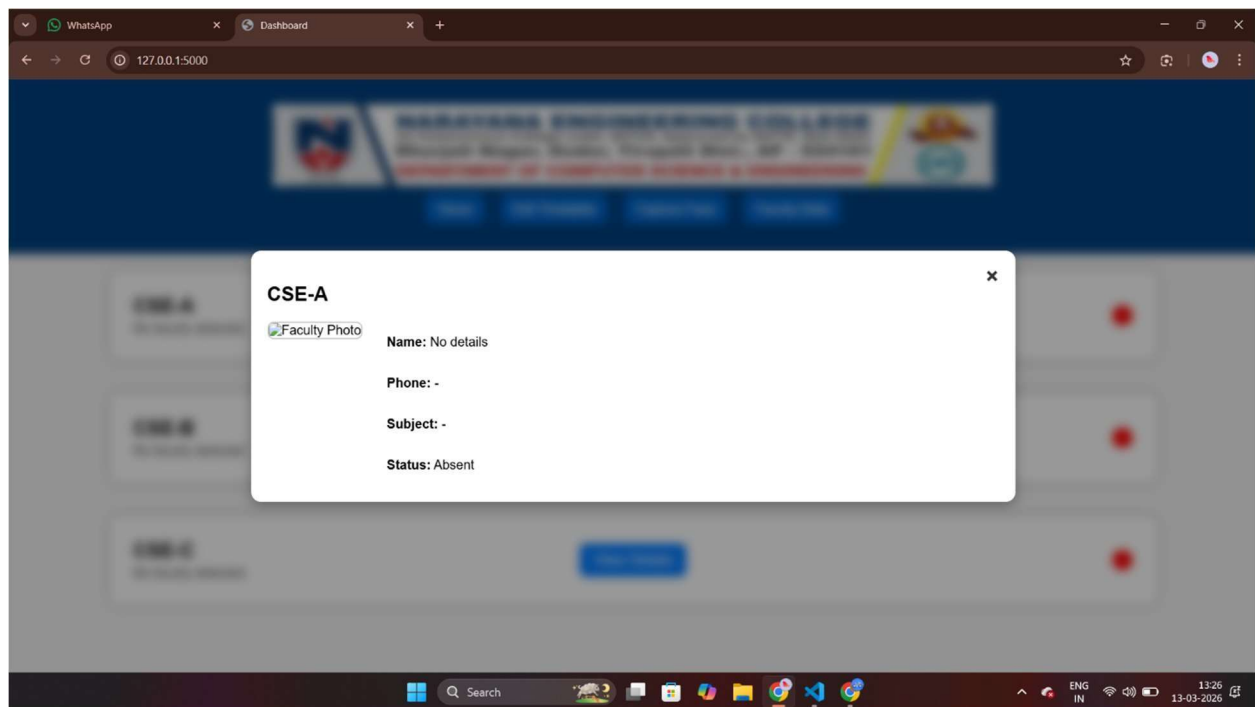


Fig-6.2.7: View Details (Before Detection)

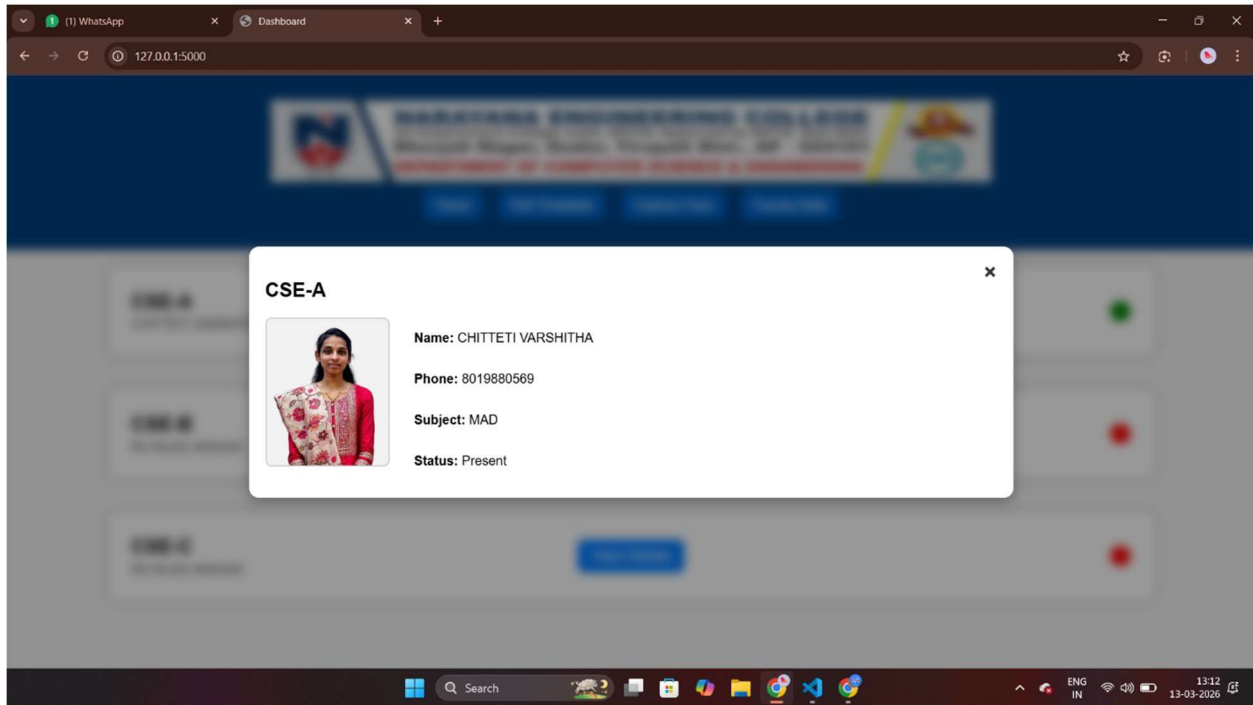


Fig-6.2.8: View Details (After Detection)

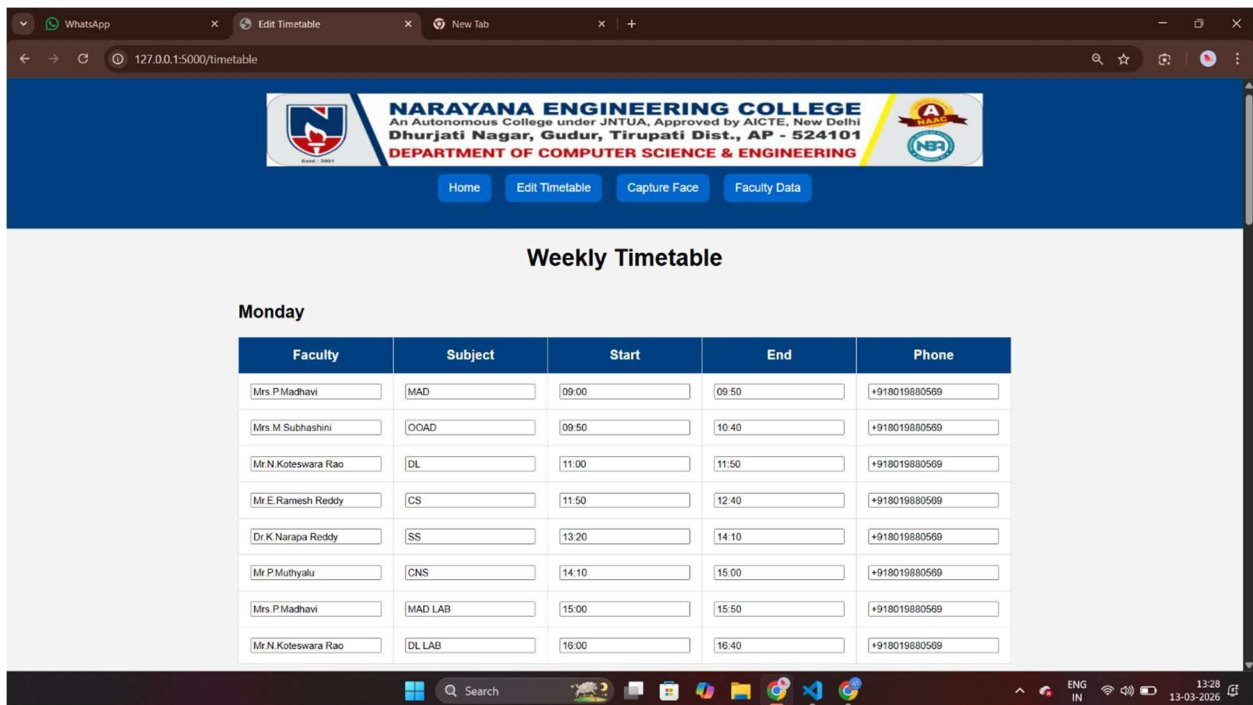


Fig-6.2.9: Timetable Menu

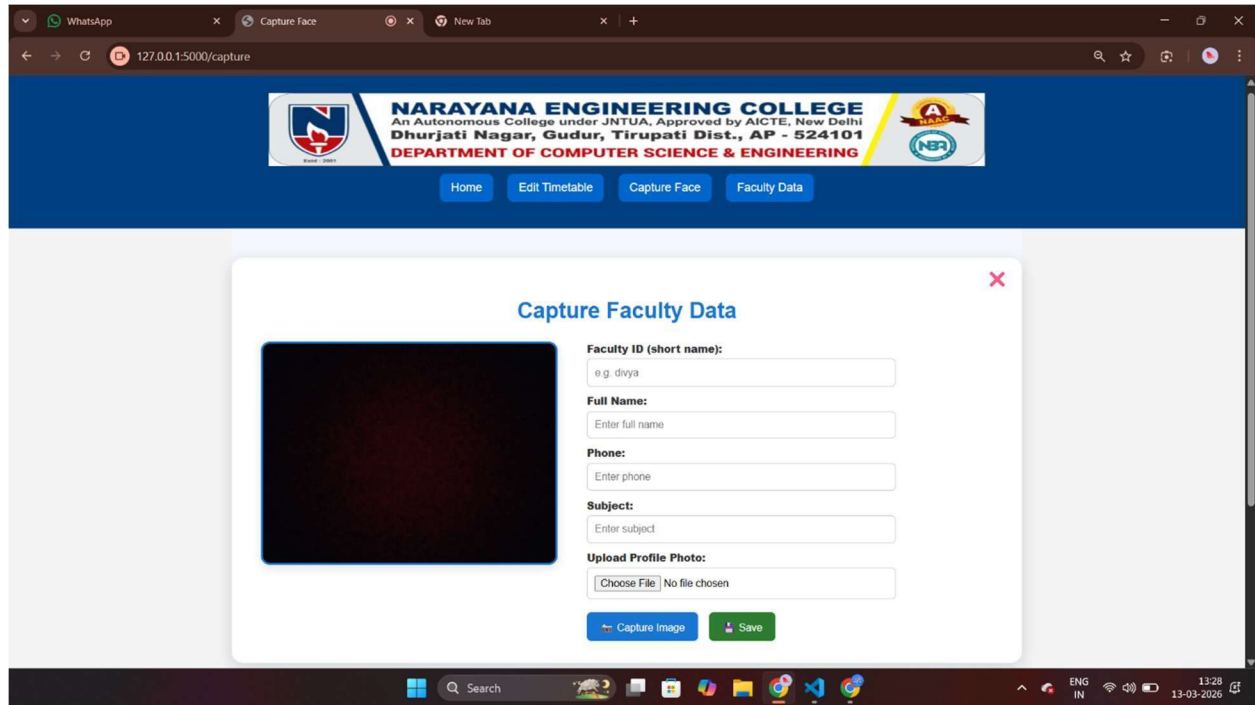


Fig-6.2.10: Capture Face Menu

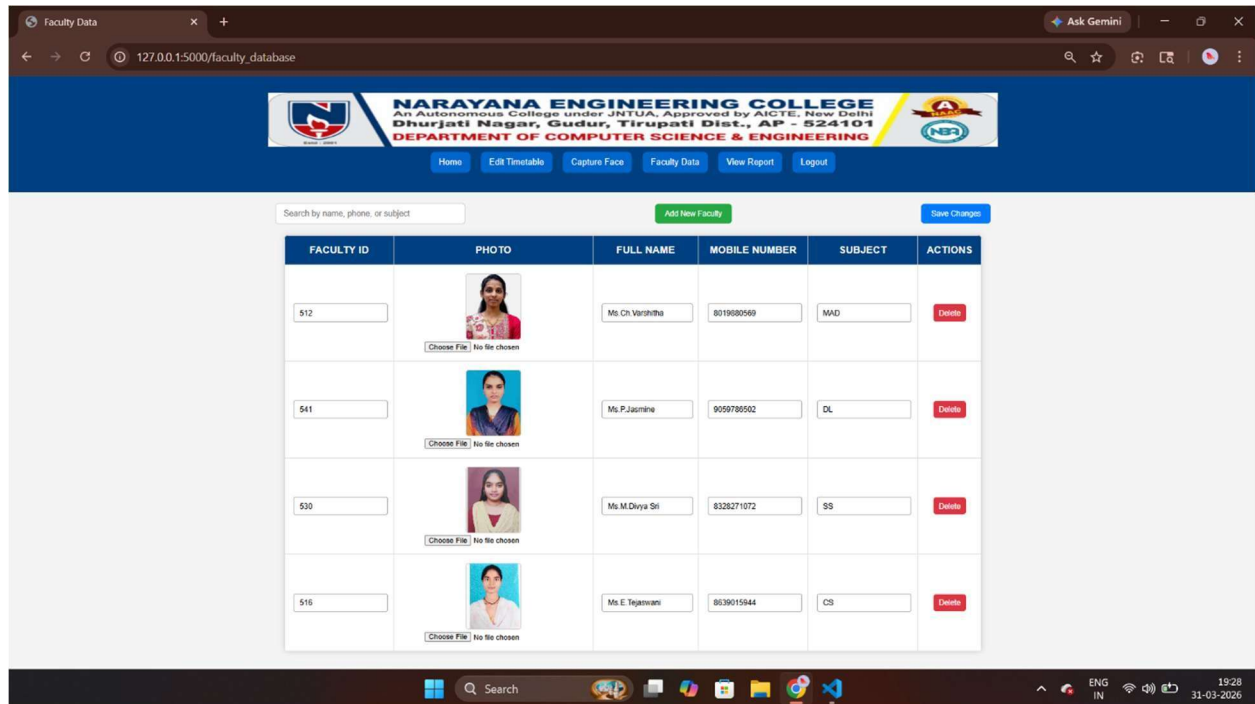


Fig-6.2.11: Faculty Data Menu

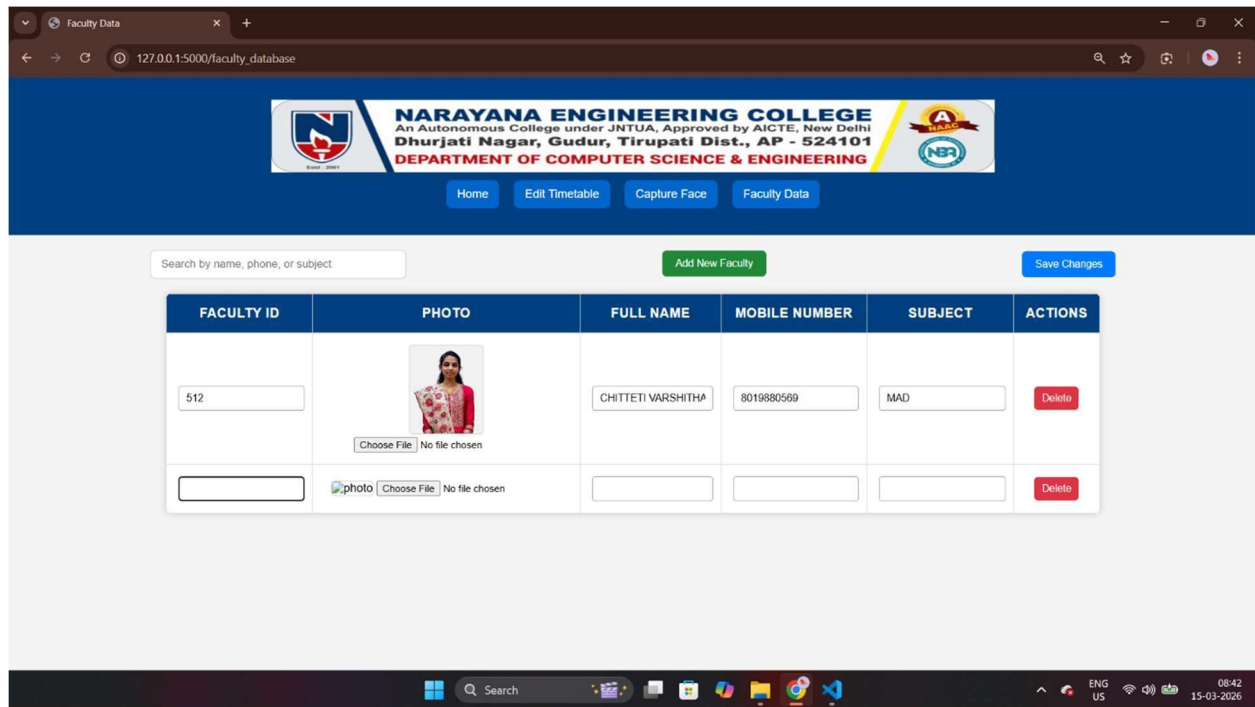


Fig 6.2.12: Add Faculty Button

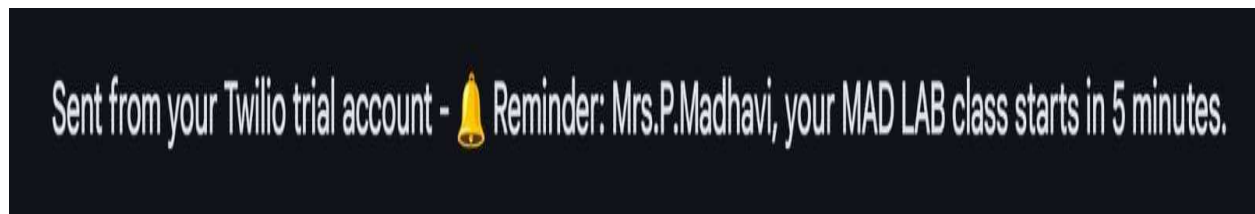


Fig 6.2.13: SMS Alert (Before 5 Minutes the Class Starts)

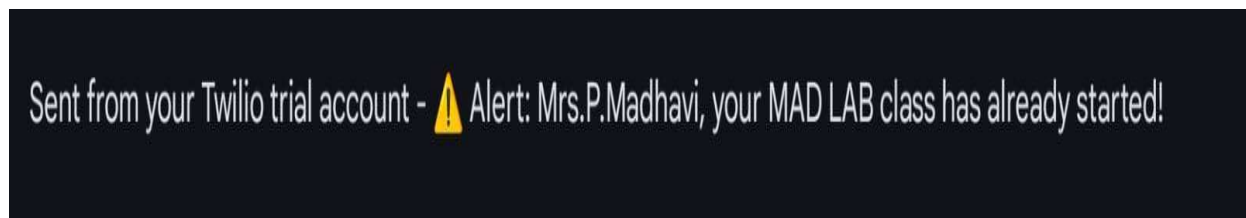


Fig 6.2.14: SMS Alert (After 5 Minutes If Faculty is Not Detected)

6.3 PERFORMANCE EVALUATION

- **Accuracy:** Face recognition achieved high accuracy under controlled environments, with minor variations due to lighting and camera angles.
- **Response Time:** The system responded quickly to real-time events, with SMS alerts triggered within 2–5 seconds.
- **Reliability:** The system showed consistent performance during testing, with no major failures.
- **Scalability:** The system can be extended to multiple classrooms, and the database efficiently handles increased user data.

6.4 COMPARATIVE ANALYSIS

Aspect	Existing Systems	Proposed System
Attendance	Manual, time-consuming	Automated, accurate using face recognition
Monitoring	Limited or offline monitoring	Real-time monitoring of classroom activities
Alerts	Not integrated	Automatic SMS alerts for faculty and students
Report Generation	Manual or periodic	Automatic weekly reports with attendance and alerts
Efficiency	Low; requires human intervention	High; reduces manual workload
Integration	Single functionality	Unified platform combining attendance, monitoring, alerts, and reporting

Aspect	Existing Systems	Proposed System
Decision Making	Human-dependent	Automated decisions based on real-time data
User Roles	No role-based access	RBAC ensures secure and role-specific access

Table-6.1: Comparative Analysis

6.5 ROLE-BASED ACCESS CONTROL (RBAC) EVALUATION

The implementation of Role-Based Access Control (RBAC) improved system security and usability:

Admin Role: Full control over dashboard, reports, and system configuration.

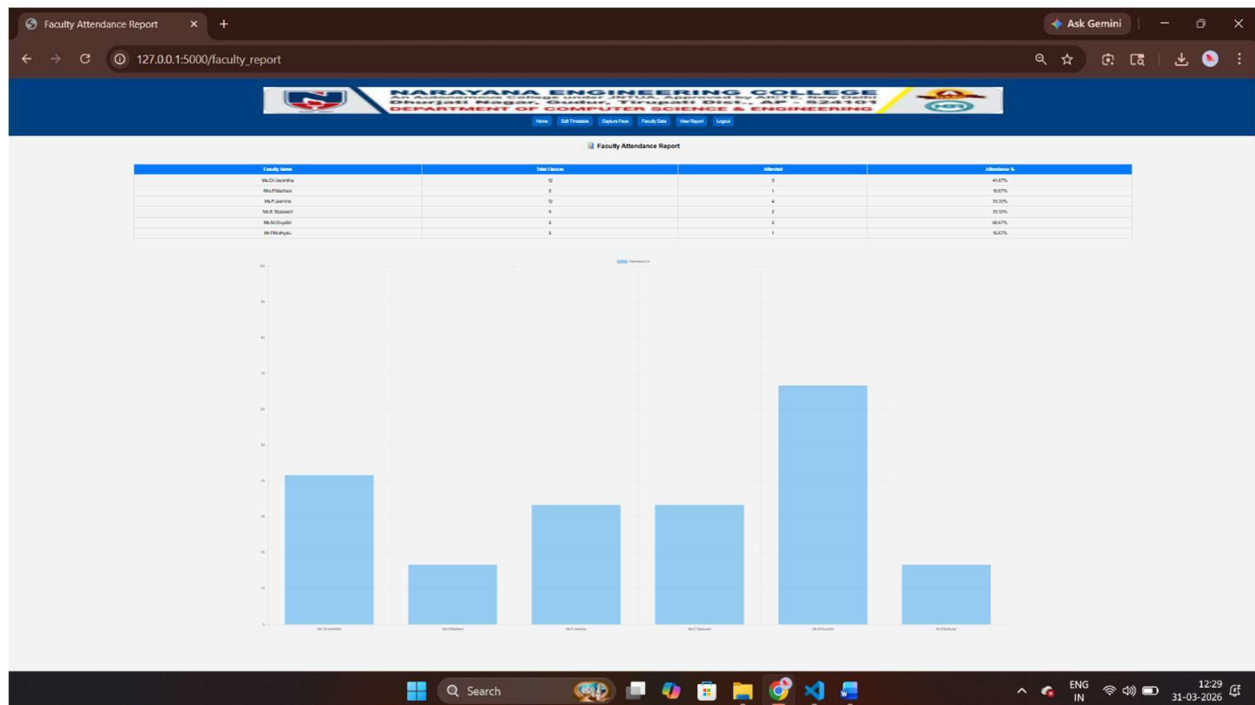


Fig-6.5.1: Admin View

Faculty Role: Limited access to reports and notifications.

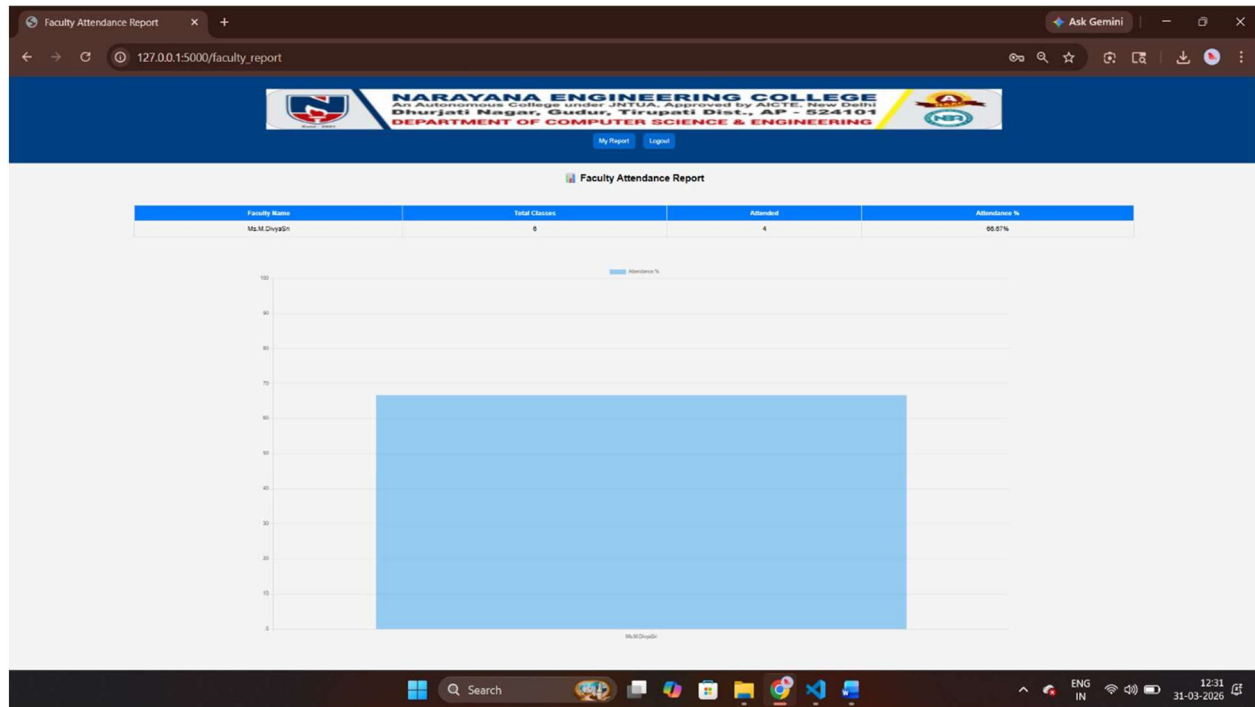


Fig-6.5.2: Faculty View

6.6 DISCUSSION

The results indicate that the Smart Classroom Monitoring System effectively addresses the limitations of traditional systems. The integration of face recognition, real-time monitoring, and alert mechanisms creates a comprehensive solution.

Key Observations:

- Automation significantly reduces manual workload.
- Real-time alerts improve classroom discipline and punctuality.
- Data-driven reports help in better decision-making.

Challenges Identified:

- Performance depends on camera quality and lighting conditions.

- Internet connectivity affects real-time features.
- Privacy concerns must be handled carefully.

6.7 FUTURE IMPROVEMENTS

The system can be enhanced in the following ways:

- Integration with AI-based behaviour analysis.
- Improved face recognition using deep learning models.
- Mobile application for better accessibility.
- Cloud-based storage for scalability.
- Enhanced privacy protection mechanisms.

6.8 CONCLUSION

The results demonstrate that the Smart Classroom Monitoring System is an effective and efficient solution for modern educational environments. It successfully integrates monitoring, recognition, alerting, and reporting into a unified system. The implementation of RBAC further strengthens security and usability. Overall, the system meets the intended objectives and provides a strong foundation for future enhancements.

7. CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION

The Smart Classroom Monitoring System presented in this project successfully addresses the limitations of traditional classroom monitoring methods by integrating modern technologies such as real-time monitoring, automated alerts, and centralized data management. The system provides an efficient and scalable solution for educational institutions to monitor classroom activities, improve discipline, and enhance overall administrative control.

One of the key achievements of this project is the development of a unified platform that connects administrators and faculty members through a structured and role-based system. The implementation of Role-Based Access Control (RBAC) ensures that different users have access only to relevant functionalities, thereby improving system security, data privacy, and usability. Administrators can manage faculty data, configure timetables, and view comprehensive reports, while faculty members receive timely notifications and can access their respective reports.

The system also improves communication efficiency by incorporating automated SMS alerts, ensuring that faculty members are informed about their schedules and responsibilities without manual intervention. This reduces the chances of miscommunication and enhances punctuality and accountability within the institution.

From a performance perspective, the system demonstrates reliability and responsiveness in handling real-time data and generating reports. The dashboard provides a clear overview of classroom activities, enabling administrators to make informed decisions quickly. Additionally, the system design ensures flexibility, allowing it to be adapted to different institutional requirements.

Overall, the project successfully meets its objectives by providing a smart, efficient, and user-friendly classroom monitoring solution. It enhances transparency, reduces manual workload, and contributes to the digital transformation of educational environments.

7.2 FUTURE SCOPE

Although the proposed system achieves its intended goals, there are several opportunities for further enhancement and expansion to make it more intelligent, scalable, and feature-rich.

One of the major improvements can be the integration of Artificial Intelligence (AI) and Machine Learning (ML) techniques. For example, face recognition can be incorporated for automated attendance tracking, and behaviour analysis can be used to monitor student engagement levels in classrooms. This would make the system more autonomous and reduce the need for manual supervision.

Another important enhancement is the development of a mobile application for both administrators and faculty members. A dedicated mobile app would provide greater accessibility, enabling users to receive notifications, view reports, and manage schedules from anywhere in real time.

The system can also be extended to include IoT (Internet of Things) devices, such as smart cameras and sensors, to collect real-time environmental and activity data from classrooms. This would improve monitoring accuracy and provide deeper insights into classroom conditions.

In terms of communication, the system can be upgraded to support multiple notification channels such as email alerts, push notifications, and integration with messaging platforms. This ensures better reach and flexibility in communication.

Another area of future work is enhancing data analytics and reporting features. Advanced analytics dashboards with visualizations, predictive insights, and performance trends can help institutions make strategic decisions based on data-driven insights.

Security can also be further strengthened by implementing advanced authentication mechanisms such as multi-factor authentication (MFA) and encryption techniques to protect sensitive institutional data.

Finally, the system can be scaled to support multi-institution deployment, allowing it to be used as a centralized platform for multiple schools or colleges.

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